Consistency theory of SMC-BVS (Wave 1 closure)

Posterior, sampler, reported-model and delivery results for $p \leq n^A$, with the Song–Liang (2015) map

For Felipe Bedoya and co-authors · compiled by the Wave 1C writer (Claude Opus 5.5) from the Wav

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For agents — how to read this chapter

- This is a *statement* document: every theorem is transcribed from the proof package named in its status box; proofs are **not** reproduced. Hypotheses are written out so that the statements can go into the paper; the proof files and the review trail are given for each result (§9).
- Status words are copied verbatim from the cited files and are never upgraded. "PROVED-CHECKED" is a Wave-1/1C ticket status: proof file + cross-family lemma checks + independent mid-point CONTINUE + exhaustive sweep fixed + two-seat council passed (both ≥ 8.0, all gates, same round) [W1C/PROTOCOL.md:35]. These ticket results are *not yet adopted by HQ into the frozen registry*; registry statuses of the State-of-Project (SoP) cards are quoted separately where they enter.
- The only new mathematics in this chapter is Lemma 7.1 (§7.1, the LS0-cap integration lemma). Its status is PROVED (by this seat), with one cross-family check (fidelity reviewer F1, Codex gpt-6.1-sol high: VERIFIED-WITH-AMENDMENTS, amendments applied, `W1C/FINAL/reviews/F1_fidelity_sol.md`); it has not been through a council and is flagged for the HQ council.
- Path aliases in citations: `W1/` = `Theory_work/Wave1/`, `W1C/` = `Theory_work/Wave1C/`, `SoP/` = `Theory_work/State_of_Project/sections/`, `CORE/` = the frozen study core `Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/`, `PROD/` = `Scripts/SMC_BVS/core/`. Every `path:line` citation was checked by `W1C/FINAL/citecheck.py` (output `W1C/FINAL/CITECHECK.txt`).

Contents

1 Setting, method and notation **3**
1.1 Data, target posterior, start . . . 3
1.2 Method v3.1 + LS0, reported model, output event . . . 3
1.3 Named implementation premises . . . 4
1.4 Three layers . . . 4

2 The Song–Liang (2015) map **4**

3 Posterior level **6**

3.1 Posterior consistency with rate (S-T, T03♯) . . . 6
3.2 Sure screening, HPM, MPM, thresholded models (C-PACK item 4) . . . 7
3.3 Misspecified candidate sets (C-MIS; Song–Liang Thm 4 / Cor 2 analog) . . . 7

4 Sampler level **8**
4.1 Unbiasedness of every bounded summary per measured round (M1(b), Lemma N1) . . . 8
4.2 What holds for the reported PIPs . . . 9

5 Reported-HPM selection consistency **10**
5.1 The route: M5(0), operational closure, LS0 . . . 10
5.2 The start-mode centre: e4 is not needed (C-PACK item 1) . . . 10
5.3 Theorem PXN: masked correlated design, p ≫ n (LH-iid) . . . 11
5.4 Theorem GEN: a general correlated class with a bounded relevance set . . . 12
5.5 Theorem GROW: growing relevance sets, including study S2 . . . 13
5.6 Reported HPM under misspecification (C-MIS Corollaries H, D, D′) . . . 14
5.7 The Song–Liang-style general-design result and study coverage . . . 14

6 Delivery **15**
6.1 The exact reduction . . . 15
6.2 Proposition F: the total tier-1 fallback . . . 16

7 Implementation status and method requirements **17**
7.1 Integration lemma: the larger LS0 cap . . . 18

8 Scope and frontier **20**

9 Provenance **21**

1 Setting, method and notation

Full definitions are in `SoP/02_method.tex` (method v3.1) and `SoP/A_notation.tex`; this section fixes only what the theorems below use.

1.1 Data, target posterior, start

Model G. Fixed (or, where stated, random) raw design $X^r \in \mathbb{R}^{n\times p}$ with nonconstant columns; $y^r = \alpha\mathbf{1} + X^r\beta + \varepsilon$, $\varepsilon \sim N_n(0, \sigma^2 I_n)$, $\sigma^2 > 0$ fixed; in random-design statements ε is independent of X^r. Columns are centred and scaled, $X = CX^r D_x^{-1}$ with $C = I - \mathbf{1}\mathbf{1}^\top/n$ and $\|x_j\|^2 = n-1$; the response is centred and scaled inside the score. $\mu := CX^r\beta = X\theta^*$, $\theta^* := D_x\beta$, $S_0 := \operatorname{supp}\beta$, $s_0 := |S_0| \geq 1$ fixed, $p = p_n \leq n^A$ for a fixed $A \geq 1$ ($p \gg n$ allowed) [SoP/04_theory.tex:70-74]; [W1C/C-GROW/STATEMENT.md:56-63].

Target (D0). $K = \min(p, \lfloor n/2 \rfloor)$, $\Omega = \{S : |S| \leq K\}$. With $G_U = X_U^\top X_U$, $z_U = X_U^\top y$, $H_U = (n+1)G_U + I$, $Q_U = y^\top y - n z_U^\top H_U^{-1} z_U$, $\nu = 0.01 + n/2$:

$$m(U) = c_n\Big[\frac{\det H_U}{\det(G_U+I)}\Big]^{-1/2}(0.02 + Q_U)^{-\nu}, \qquad w(k) = \Big[2\binom{2p-1}{k}\Big]^{-1}, \qquad \pi(S) \propto w(|S|)\, m(S) \;\; (S \in \Omega),$$

the shipped $g = n$ ridge score with IG(0.01, 0.01) and the capped BB$(1, p)$ size prior [SoP/04_theory.tex:71-72]. $F(S) := \log w(|S|) + \log m(S)$. $\mathrm{PIP}_j := \pi(j \in S)$, $\mathrm{MPM} := \{j : \mathrm{PIP}_j > \frac{1}{2}\}$, $\xi_q := \{j : \mathrm{PIP}_j > q\}$, and the *exact-posterior* $\mathrm{HPM} := \arg\max_\Omega \pi$ (distinct from the method's reported model $\hat{S}^+$ below) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:11-18].

Blocks, start law, start mode. The method partitions the columns into blocks $C_1, \ldots, C_J$ (complete linkage on $1 - |\hat{r}|$, top-down cut at $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$, next-fit packing; a deterministic function of X). Block tables $r_b(U) = m(U)/m(\emptyset)$; start $\gamma_0(S) \propto w(|S|)\prod_b r_b(S \cap C_b)$ on Ω, $\pi_0 = \gamma_0/\bar{Z}_0$; the **start mode** $\tilde{S}_0 = \arg\max_\Omega \gamma_0$ is computed exactly by a block DP (`CORE/cpp/v3_start.h`) [W1/W1-GEN/STATEMENT.md:26-38].

1.2 Method v3.1 + LS0, reported model, output event

Local search (M §6.10). From a centre, scan every feasible flip (additions beyond K excluded) and every swap with the centre frozen; every scanned model is scored; move to the best scanned neighbour if its F is *strictly* larger than the centre's (exact F ties among neighbours broken lexicographically); at most ℓ moves, then one further scan; `local_limit` iff that scan still improves; `local_certified` iff the returned model is a scanned centre with no improving neighbour [SoP/02_method.tex:463-467]; code [CORE/cpp/v3_engine.h:376-425]. Production cap $\ell_n = \lceil \log_2(n + p + 1) \rceil$.

LS0. After the final local search (and after the extra search from c_4 when $R' > 4$), one local search with the §6.10 rules from the start mode $\tilde{S}_0$; LS0 reads no uniforms, every model it scores joins the all-scored set and the stores, its centre is scored (emitted); a completed LS0 capped at `local_limit` still returns output [SoP/02_method.tex:490-501]. LS0 is **not** in the frozen production core (which calls only the search from the global best, [PROD/cpp/v3_core.cpp:447]); it is merged on branch `w1c/ls0-merge` by C-ENG (§7).

Reported HPM. $\hat{S}^+$ is the maximum of the all-scored set $\mathcal{S}^+$ (every model ever scored, including LS0) for the strict total order $(F(S), -|S|, \text{lex})$ [SoP/02_method.tex:468-473].

Output event and potential run. $\mathcal{O}$ is the event that the fit returns estimates (status `exact` or `completed`); $U_n := \mathcal{O}^c$. Guarantees are stated for the potential uninterrupted run and as $\mathbb{P}(\mathcal{O} \cap \cdot)$, never as $\mathbb{P}(\cdot \mid \mathcal{O})$ [SoP/02_method.tex:579-583]. Bars denote potential-run objects; the never-scored term is

$$D_n := \mathbb{P}\big(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\}\big) \quad \text{(potential-run notation; the bar is not a complement)}$$

[SoP/04_theory.tex:84].

1.3 Named implementation premises

The reported-model theorems are proved for the method *as specified* under named premises; these are not model assumptions and are not discharged by the theorems [SoP/04_theory.tex:84]: **EX-A** exact scores and arithmetic (an idealisation, not a validated floating-point property); **EX-S** fresh independent ideal uniforms; **CAL** a successful frozen calibration record on streams 0–9; **IND-T** final streams independent, $R' = 4k$ declared before stream 10 (needed when $R' > 4$); **GUARD-W** stop-only guard (operational interruption ⇒ no output); **SR** the stop/return rule; **OP-LS0** LS0 integrated as specified; **e4** the emitted LS0 centre is the prescribed start-mode maximiser (replaced by the weaker **e4′**, §5.2); **complete scans**. The delivery results (§6) add the idealisation premises **IDEAL-W** (unbounded words), **MACH** (wall time and bytes proportional to operations and words; allocations succeed within the declared memory; no external interruption), **CONFIG** (default declared list, polynomial certificate budgets, fixed worker count) and **TDP-IMPL** (the e4 tie fallback is W1-N3's reference algorithm TDP), and do *not* assume CAL [W1C/C-DLV/STATEMENT.md:21-46].

1.4 Three layers

As in the SoP regime map [SoP/04_theory.tex:822], results live on three layers that compose only explicitly: **(i)** the exact posterior π concentrates on S_0 (§3); **(ii)** the sampler's outputs track π or $h(S_0)$ (§4); **(iii)** the reported model $\hat{S}^+$ equals S_0 jointly with output (§5). Delivery, $\mathbb{P}(\mathcal{O}) \to 1$, is a fourth question (§6). M5(0) composes (i) and (iii) at every n:

$$\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2\,\mathbb{E}_0[1 - \pi(S_0)] + D_n$$

(the deterministic part: if $S_0 \in \mathcal{S}^+$ and $1 - \pi(S_0) < \frac{1}{2}$ then $\hat{S}^+ = S_0$), status T38-(0) "PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))" [SoP/04_theory.tex:477-489]. Hence every reported-HPM theorem below reduces to $\mathbb{E}_0[1 - \pi(S_0)] \to 0$ (layer (i)) plus $D_n \to 0$ (discovery of S_0 by the search).

2 The Song–Liang (2015) map

Song & Liang (2015, JRSS-B 77(5), 947–972) is the template for the story: posterior consistency, sure screening, thresholded and MAP-model consistency, and a misspecification theorem. The table updates C-PACK's map v2 [W1C/C-PACK/SONG_LIANG_MAP.md:1-12] with C-MIS, C-GROW and C-DLV/A1. Two global differences: (a) their correctly specified asymptotics also assume $P_n/n^\theta \to \infty$ for some $\theta > 0$, while ours impose $p_n \leq n^A$ (LH-iid: also $p_n \to \infty$) with no positive-power lower growth bound — different domains, not identified [W1C/C-PACK/SONG_LIANG_MAP.md:10-12]; (b) their Theorem S is in an online supplement that was **NOT INSPECTED** (publisher access failed); comparison is with the main-text description only [W1C/C-PACK/REPORT.md:87] .

Song–Liang	their content	our analog (this chapter)	status (verbatim)
Thm 1 (p. 952), posterior consistency	Hellinger contraction, exponential inner/outer rates, $\|x_i\| \le 1$, approximate sparsity	**No Hellinger/predictive analog.** Selection analog: Thm 3.1 (S-T/T03♯), $1-\pi(S_0) = O_P(\sqrt{\log n/n})$, fixed design, $p \le n^A$	T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2); Hellinger: CJ20 FW20 Not delivered; selection counterpart in S-T only
Thm 2 (p. 953)	random design, Condition 1, strict sparsity; Hellinger	random-design *selection* analogs: PXN (P) on LH-iid (joint law); GEN class via (C4), instance I-c (Cor. 3.3)	W1-PXN, W1-GEN PROVED-CHECKED (Wave 1, not adopted by HQ)
Cor 1 (p. 954)	$P_n < \exp(Cn^\alpha)$, two prior branches, rates	rate $\sqrt{\log n/n}$ for $p \le n^A$ only; exponential p not covered	CJ18 No completed prior-tail/contraction route; p ≤ n^A stays in scope
Cond 2, (16) (p. 954)	second moments ⇒ convergence in probability with exponential inner threshold	all our statements are O_P with a polynomial rate; no exponential or expectation rate	— (comparison of modes)
Thm 3, sure screening (p. 956)	per true predictor, $\mathbb{P}^*[\pi(t \in \xi) < 1 - e^{-c_1 n\varepsilon_n^2}] < e^{-c_1 n\varepsilon_n^2}$ (sensitivity only)	SL0(i), SL1(a): $\max_j \lvert\mathrm{PIP}_j - 1\{j \in S_0\}\rvert \le 1 - \pi(S_0) = O_P(\sqrt{\log n/n})$ — sensitivity *and* specificity, uniformly in j; SL1′ without beta-min (large signals, sensitivity only)	PROVED (by this seat, unchecked); R34 re-derived; C-PACK council passed
ξ_q (p. 956)	$\mathbb{P}^*(t \subseteq \xi_q) \to 1$	SL1(b): $\mathbb{P}(\xi_{q_n} = S_0) \to 1$ (equality) for fixed q or $r_n = o(\min(q_n, 1-q_n))$; exact MPM consistent	as row above
Thm S, MAP (supplement)	"establishes the consistency of MAP models" — exact hypotheses **NOT INSPECTED**	exact HPM: SL0(iv), SL1(c). **Reported** $\hat{S}^+$, jointly with output: LH-iid (Thm 5.2), GEN class (Thm 5.3), growing relevance sets incl. study S2 (Thm 5.4), misspecified candidate sets (Cor. 5.6); unconditionally $\mathbb{P}(\hat{S}^+ = S_0) \to 1$ for the patched calibration on LH, LH-iid, GEN class (Prop. 6.3)	W1-PXN, W1-GEN, C-GROW, C-MIS PROVED-CHECKED; C-DLV PARTIAL (Prop. F: PROVED-CHECKED for the patched calibration law); A1 HOLDS-WITH-AMENDMENTS
SAM inclusion probabilities (§3.2, §7)	importance-weighted estimates; not itself Thm 3	islandwise weighted bound (Cor. 4.2); stacking route M1(i); store-restricted SL2. **No in-probability sure-screening theorem for the shipped pooled $R' = 4$ PIPs**	Cor. E PROVED under the same premises; M1(i) (h), (i): PROVED (by this seat, unchecked)
Cond 3, identifiability (p. 955)	uniform exponential-moment identifiability	$\mathrm{SE}_\lambda/\mathrm{SE}_w$ + beta-min; GEN's Gram floors; GROW's (G2)–(G3) — related role, *not* asserted equivalent	— (comparison)
Thm 4, misspecified subset (p. 957)	fixed-size candidate set s, Condition 4: density consistency at the KL projection f_0, a.s.	Thm 3.5: *exact-model* concentration $\pi_{\mathcal{C}}(S_0^{\mathcal{C}}) \to 1$ at rate $\sqrt{\log n/n}$, $p_{\mathcal{C}} \le n^A$, under the Gaussian-residual condition (GR) (stronger hypotheses); Thm F (fixed design, exact projection). No density analog	C-MIS PROVED-CHECKED
Cor 2, (19) (p. 957)	$\pi(\tilde{s} \subseteq \xi \mid D_n^s) \to_P 1$, $\tilde{s} = \operatorname{supp}\beta_0$ (screening); withdrawn under a binding cap	MIS no-beta-min clause; Lemma GD(a) recovers Cor. 2's conclusion in the jointly Gaussian fixed-$p_{\mathcal{C}}$ case, with a rate; Prop. P neighbour rule	C-MIS PROVED-CHECKED

Song–Liang	their content	our analog (this chapter)	status (verbatim)
Cond 5, $\lvert\mathrm{cov}(x_t, y)\rvert > 0$ (p. 958)	keeps every true predictor in SAM's screened set	analogy only: GEN's certificate (C1′) and GROW's block visibility (TV) are block-level conditions; masked truths (zero block coefficient) are recovered by the descent (GEN)	GEN PROVED-CHECKED; (C1′) VERIFIED-WITH-AMENDMENTS (R32)
§6, time (p. 968)	parallel time with $K_n \asymp \sqrt{P_n}$ processors	M8 accounting remark only; no speed claim. Delivery under a declared polynomial allocation: Prop. 6.3	M8 assembly PROOF SKETCH (by this seat, unchecked)

Sources: [W1C/C-PACK/SONG_LIANG_MAP.md:14-28]; [W1C/C-MIS/STATEMENT.md:15-35]; [W1C/C-MIS/REPORT.md:3-9]; [W1C/C-GROW/REPORT.md:24-30]; [W1C/HQ_AUDIT/A1_propF_redteam.md:3].

Plain reading — the map in one paragraph

Every correctly specified Song–Liang *selection* result (Thm 3, ξ_q, MAP) has an exact-posterior analog that is a one-line consequence of $1-\pi(S_0) \to 0$ (Lemma SL0, deterministic), with rate $\sqrt{\log n/n}$ for $p \le n^A$ and with specificity added. Beyond Song–Liang, the method's *reported* model is proved consistent jointly with output on a general correlated class (GEN), on growing relevance sets including the study's AR(1) design (GROW), and under misspecification (C-MIS Cor. D), and — with a one-line calibration fallback — unconditionally (Prop. F). Gaps: Hellinger/predictive contraction and exponential rates; exponential p; shipped pooled $R' = 4$ PIPs; Theorem S's hypotheses not inspected; density-level misspecification.

3 Posterior level

3.1 Posterior consistency with rate (S-T, T03♯)

Theorem 3.1 (S-T / T03♯; card TH09). *Fixed-design triangular array (model G), fixed $s_0 \ge 1$, $p \le n^A$ for a fixed A, $s_0 \le K$ eventually. Assume*

- ***R**: $\mu = X\theta^*$, $\|\mu\|^2 \le Mn$, and $\lambda_{\min}(\hat{R}_{S_0}) \ge \phi_0^2 > 0$ ($\hat{R} = X^\top X/(n-1)$);*
- ***tail cutoff**: $R_m = \sum_{k=\max(m+1,0)}^{K} \binom{p}{k} w(k)/w(s_0)$, $m_n^\sharp = \min\{m \ge s_0 : R_m \le n^{-(s_0+2)/2}\}$ (deterministic, data-free);*
- ***SE**$_\lambda(j_0)$, $j_0 = \lceil (2A+1)s_0 \rceil + 1$: every $S_0 \cup J$, $|J| \le j_0$, has normalised minimum eigenvalue $\ge \phi_0^2$; **SE**$_w(m_n^\sharp; g_0)$: every $S_0 \cup J$, $|J| \le m_n^\sharp$, has unnormalised minimum eigenvalue $\ge g_0 > 0$;*
- ***beta-min**: BM-∞, $\theta_{\min}^{*2} n\phi_0^2/(\sigma^2 \log n) \to \infty$; or BM($c$) with a fixed $c > c_0(s_0, A)$.*

Then $1-\pi(S_0) = O_P(\sqrt{\log n/n})$; the exact HPM and MPM are consistent. Without beta-min, for $J_n(c) = \{j \in S_0 : \theta_j^{*2} \ge c\sigma^2 \log n/(n\phi_0^2)\}$, $c > c_0$: $\pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$.

Status and evidence

T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2). T02(i♯), T02(ii♯): PROVED; SF/XF VERIFIED. T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY [SoP/04_theory.tex:765]. Statement [SoP/04_theory.tex:774-782]. Algorithm-free.

Scope. "No expectation rate inferred from O_P, no universal onset n_0, no random-design or growing-sparsity extension" (card Repairs row); $p \le n^A$ "comes from the prior and proof technique; e^{Cn^α} OPEN". The binding condition of the exhibited families first holds near $n \approx 2 \cdot 10^6$ — an illustration for those families, not a universal onset [SoP/04_theory.tex:787-811].

Random designs. T03♯ is a fixed-design theorem. Random-design versions used below: (a) W1-PXN (P) on LH-iid, $1-\pi(S_0) = O_P(\sqrt{\log n/n})$ under the joint law of (X^r, ε), via T03♯ along design sequences plus a design event and de-randomisation (XF VERIFIED, R19) [W1/W1-PXN/REPORT.md:49]; (b) GEN's random-design class (C4): T03♯ along every deterministic sequence in the X-events with common constants and envelopes, plus the diagonal tightness transfer (instance I-c, [W1C/C-PACK/lemmas/I4_SL_corollaries.md:57-60]). Conditional convergence alone would not suffice; the every-sequence/common-envelope quantifiers are needed.

3.2 Sure screening, HPM, MPM, thresholded models (C-PACK item 4)

Lemma 3.2 (SL0, deterministic; every data set, every n). *Let $\delta := 1-\pi(S_0)$, with $\pi(S_0) := 0$ if $s_0 > K$. Then: (i) $|\mathrm{PIP}_j - 1\{j \in S_0\}| \le \delta$ for every j, hence $\max_{j\in S_0}(1-\mathrm{PIP}_j) + \max_{j\notin S_0}\mathrm{PIP}_j \le 2\delta$; (ii) $|\pi(A \subseteq S) - 1\{A \subseteq S_0\}| \le \delta$ for every A, and if $S_0 \in \Omega$, $|\pi(h) - h(S_0)| \le \delta\,\mathrm{osc}(h)$ and $\sup_{h:\Omega\to[0,1]}|\pi(h)-h(S_0)| = \delta$; (iii) $\delta < \min(q, 1-q) \Rightarrow \xi_q = S_0$, in particular $\delta < \frac{1}{2} \Rightarrow \mathrm{MPM} = S_0$; (iv) $\delta < \frac{1}{2} \Rightarrow S_0$ is the unique maximiser of π; (v) $\pi(|S| = s_0) \ge 1-\delta$. (vi) The constant 2 in (i) is sharp over arbitrary laws on Ω (two-point law); attainment by a D0 posterior is* not *claimed.*

Corollary 3.3 (SL1). *If $\delta_n = O_P(r_n)$, $r_n \to 0$, under some law: (a) $\max_j|\mathrm{PIP}_j - 1\{j\in S_0\}| = O_P(r_n)$ and $\mathbb{E}\max_j|\cdot| \le \mathbb{E}\delta_n \to 0$ (no expectation rate); (b) $\mathbb{P}(\xi_{q_n} = S_0) \to 1$ whenever $r_n = o(\min(q_n, 1-q_n))$, in particular for fixed q, and $\mathbb{P}(\mathrm{MPM} = S_0) \to 1$; (c) $\mathbb{P}(\mathrm{HPM} = S_0) \ge \mathbb{P}(\delta_n < \frac{1}{2}) \to 1$, $\mathbb{P}(\mathrm{HPM} \ne S_0) \le 2\mathbb{E}\delta_n$; (d) $\sup_{h:\Omega\to[0,1]}|\pi(h)-h(S_0)| = O_P(r_n)$; (e) $\pi(|S| = s_0) = 1 - O_P(r_n)$. Instances with $r_n = \sqrt{\log n/n}$: **I-a** all T03♯ hypotheses (fixed design); **I-b** all W1-PXN (P) hypotheses (LH-iid, joint law); **I-c** GEN (C4) (joint law); **I-d** LH exact Gram (Thm5-LH keeps SAME-FAMILY-ONLY).*

Corollary 3.4 (SL1′: no beta-min, large signals only). *Retaining T03♯'s other hypotheses, $\max_{j\in J_n(c)}(1-\mathrm{PIP}_j) \le \pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$ ($= 0$ if $J_n(c) = \emptyset$). No specificity or full-support recovery follows from this clause alone.*

Status and evidence

SL0, SL1, SL1′: PROVED (by this seat, unchecked), independently re-derived by R34; v2 re-read by R38 (mid-point CONTINUE) and R39 (sweep); C-PACK council round 1 passed (Sol 9.0, Opus 8.7) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:3-7]. Statements [W1C/C-PACK/lemmas/I4_SL_corollaries.md:20-48]; [W1C/C-PACK/lemmas/I4_SL_corollaries.md:53-68]. Falsification: 216 exact D0 posteriors, 84,096 pairs, 0 violations (`W1C/C-PACK/numerics/R34_falsify_out.txt`).

3.3 Misspecified candidate sets (C-MIS; Song–Liang Thm 4 / Cor 2 analog)

Setting. A candidate set $\mathcal{C}$, $|\mathcal{C}| = p_{\mathcal{C}} \le n^A$; the method is *re-run on the candidate data* $(X^r_{\mathcal{C}}, y^r)$ with the shipped score and prior ($\mathrm{BB}(1, p_{\mathcal{C}})$, $K = \min(p_{\mathcal{C}}, \lfloor n/2\rfloor)$) — not obtained by conditioning the full-p posterior. True model $y^r = \alpha\mathbf{1} + X^r_{\mathcal{C}}\beta_{\mathcal{C}} + \omega + \varepsilon$, $\omega = X^r_{\mathcal{M}}\beta_{\mathcal{M}}$ the omitted contribution [W1C/C-MIS/STATEMENT.md:37-50].

(GR) Gaussian residual. Conditionally on $X^r_{\mathcal{C}}$, $\omega = a\mathbf{1} + X^r_{\mathcal{C}}\gamma + \xi$ with $\xi \sim N_n(0, \tau^2 I_n)$, (a, γ) measurable in $X^r_{\mathcal{C}}$, $\tau^2 \ge 0$ **fixed**, ξ independent of ε; the method's uniforms independent of everything. Define $\beta^{\mathcal{C}} := \beta_{\mathcal{C}} + \gamma$, $S_0^{\mathcal{C}} := \mathrm{supp}\,\beta^{\mathcal{C}}$, $s_0^{\mathcal{C}} := |S_0^{\mathcal{C}}|$, $\sigma^2_{\mathcal{C}} := \sigma^2 + \tau^2$. Under GR the candidate-only problem *is* a well-specified Gaussian linear model with truth $\beta^{\mathcal{C}}$ and noise $\sigma^2_{\mathcal{C}}$ [W1C/C-MIS/STATEMENT.md:54-58].

(H-MIS) = the T03♯ hypotheses for the candidate problem with $(p, S_0, s_0, \theta^*, \mu, \sigma^2) \mapsto (p_{\mathcal{C}}, S_0^{\mathcal{C}}, s_0^{\mathcal{C}}, \theta^{\mathcal{C}}, \mu^{\mathcal{C}}, \sigma^2_{\mathcal{C}})$, in particular $s_0^{\mathcal{C}} \ge 1$ fixed and beta-min relative to $\sigma^2_{\mathcal{C}}$.

Theorem 3.5 (MIS). *(i)* Fixed candidate design, *GR and (H-MIS):* $1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y) = O_P(\sqrt{\log n/n})$; $\max_j|\mathrm{PIP}_j - 1\{j\in S_0^{\mathcal{C}}\}| \le 1-\pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)$; $\mathbb{P}(\mathrm{MPM}_{\mathcal{C}} \ne S_0^{\mathcal{C}}) \to 0$; $\mathbb{P}(S_{\mathcal{C}}^{\dagger} \ne S_0^{\mathcal{C}}) \le 2\mathbb{E}[1-\pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)] \to 0$; without beta-min, $\pi_{\mathcal{C}}(S \not\supseteq J_n^{\mathcal{C}}(c) \mid y) = O_P(\sqrt{\log n/n})$ for fixed $c > c_0$. Consequences: a* proxy

($j \in S_0^{\mathcal{C}} \setminus S_0$) has $\mathrm{PIP}_j \to 1$*; a* cancelled *true candidate ($\beta_j + \gamma_j = 0$) has* $\mathrm{PIP}_j \to 0$*.* ***Corollary N****: every $T \neq S_0^{\mathcal{C}}$ has $\pi_{\mathcal{C}}(T \mid y) \leq 1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)$; in particular the retained truth $S_0 \cap \mathcal{C}$ and the sample-projection support lose all mass whenever they differ from $S_0^{\mathcal{C}}$.*
(ii) Random candidate design, envelope form*: the same conclusions under the joint law, if on measurable design sets $\mathcal{A}_n$ with $\mathbb{P}(\mathcal{A}_n) \to 1$ every $x \in \mathcal{A}_n$ satisfies the finite-n inequalities of (H-MIS) with fixed constants, $s_0^{\mathcal{C}}(x) = s \geq 1$ and the beta-min envelope (diagonal argument, Lemma DR); (ii′) without beta-min, the inclusion clause.*
(iii) Jointly Gaussian rows *($\mathcal{M} := S_0 \setminus \mathcal{C}$, rows iid $N(m, \Sigma)$, $\Sigma_{\mathcal{C}\mathcal{C}} \succ 0$): at each n GR holds with $\gamma = \Sigma_{\mathcal{C}\mathcal{C}}^{-1}\Sigma_{\mathcal{C}\mathcal{M}}\beta_{\mathcal{M}}$, $\tau^2 = \beta_{\mathcal{M}}^{\top}(\Sigma_{\mathcal{M}\mathcal{M}} - \Sigma_{\mathcal{M}\mathcal{C}}\Sigma_{\mathcal{C}\mathcal{C}}^{-1}\Sigma_{\mathcal{C}\mathcal{M}})\beta_{\mathcal{M}}$; $\beta^{\mathcal{C}}$ is the population slope with intercept (= Song–Liang's β_0 and $S_0^{\mathcal{C}} = \tilde{s}$ for centred rows). MIS (i)–(ii) apply only when in addition $\tau_n^2 \equiv \tau^2$ is fixed.* ***Neighbour rule*** *(Prop. P(d)): with one active omitted variable k and Θ the precision of the marginal vector $(x_{\mathcal{C}}, x_k)$, a candidate that is null in the full model belongs to $S_0^{\mathcal{C}}$ iff $\Theta_{jk} \neq 0$ — partial, not marginal, correlation decides (AR(1): the two neighbours; a one-factor design can make $s_0^{\mathcal{C}}$ grow, outside the theorem).* ***Lemma GD****: (a) for fixed $p_{\mathcal{C}}, m, \Sigma, \beta$ fixed, $\beta^{\mathcal{C}} \neq 0$, (ii) applies and the inclusion clause recovers Song–Liang's Cor. 2 conclusion in the jointly Gaussian case, now with a rate; (b) growing $p_{\mathcal{C}} \leq n^A$ under stated population restricted-eigenvalue, variance, energy and projected beta-min assumptions — projection sparsity is* assumed*.*

Theorem 3.6 (F: fixed design, exact sparse projection)**.** *In contrast coordinates, assume (EP): $r := \mu_c - X_{c,\mathcal{C}}\theta^{\mathcal{C}}$ is orthogonal to every candidate,* $\mathrm{supp}\,\theta^{\mathcal{C}} = S_0^{\mathcal{C}}$*; set $\Delta_n = \|r\|^2/N$, $\sigma_{\mathcal{C},n}^2 = \sigma^2 + \Delta_n$. Under $s_0^{\mathcal{C}} \geq 1$ fixed, $p_{\mathcal{C}} \leq n^A$, G, $R^{\mathcal{C}}$, $SE_\lambda(j_0)$, $SE_w(m_n; g_0)$ anchored at $S_0^{\mathcal{C}}$ (cutoff $s_0^{\mathcal{C}} \leq m_n \leq \min(K, C\log n)$), BM-∞ or BM($c > c_0$) relative to $\sigma_{\mathcal{C},n}^2$, and the size tail* ***TAIL***$^{\mathcal{C}}(m_n)$*: $\pi_{\mathcal{C}}(|S| > m_n \mid y) = O_P(\sqrt{\log n/n})$ — the conclusions of T03♯ hold for $S_0^{\mathcal{C}}$. TAIL$^{\mathcal{C}}$ is discharged (a) for bounded $p_{\mathcal{C}}$ (Cor. F(a)), and (b) under (TL) $\sigma^2/(\Delta_n + \sigma^2(1 - d_n/N)) \leq \frac{3}{4}(1 - \kappa)$ eventually, $d_n = \mathrm{rank}\,X_{c,\mathcal{C}}$ (Lemma T, Cor. F(b)).* ***Lemma Z*** *(null projection, fixed candidate design): if $s_0^{\mathcal{C}} = 0$, $p_{\mathcal{C}}$ is bounded, and $\lambda_{\min}(X_{\mathcal{C}}^{\top}X_{\mathcal{C}}) \geq g_0 > 0$ eventually, then under GR, or under EP with $\theta^{\mathcal{C}} = 0$, $1 - \pi_{\mathcal{C}}(\emptyset \mid y) = O_P(n^{-1/2})$. The general growing-p null case is not claimed.*

Status and evidence

C-MIS: PROVED-CHECKED (narrowed form): Theorem MIS (i)–(iii) with Prop. P, Lemma DR, Lemma GD(a)/(b); Corollaries H, D, D′; Theorem F with F(a), F(b), Lemma T; Lemma Z; Corollary N; Examples E1–E3. Statement lock R01; lemma checks R02, R03, R05; mid-point R04 CONTINUE; sweep R06 (0 fatal); council round 1 Sol 9.2 / Opus 9.0, all gates PASS [W1C/C-MIS/REPORT.md:21-29]. Statements [W1C/C-MIS/STATEMENT.md:64-95]; [W1C/C-MIS/STATEMENT.md:126-141]; [W1C/C-MIS/STATEMENT.md:101-105]. Inputs: T03♯ as above; T38-(0) PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5)).

Boundary. Example E1: x_1, x_2, x_3, z iid $N(0,1)$, omitted $x_k = -x_1 + x_2 + z$, $y = x_1 + x_k + \varepsilon$: $S_0^{\mathcal{C}} = \{2\}$ (proxy), the true candidate 1 is cancelled, and both $\{1\}$ and the dense sample-projection support lose their mass. E2: a common random omitted slope with Gaussian marginals defeats the transfer (GR's covariance content matters). E3: EP cannot be dropped for a pre-named target. **Open** (§8): EP with growing $p_{\mathcal{C}}$ and small residual $0 < \Delta_n \lesssim \sigma^2(1/3 + d_n/N)$; approximate projection; outside GR; random-design EP [W1C/C-MIS/REPORT.md:131-139].

4 Sampler level

4.1 Unbiasedness of every bounded summary per measured round (M1(b), Lemma N1)

Island r runs on final stream $9 + r$ after a calibration record $\mathcal{C}$ frozen on streams 0–9; $\hat{Z}_r$ is its evidence estimate, Z the normalising constant, $W_r := \hat{Z}_r/Z$. After each of the four measured `population_round(P,1.0)` calls the population is $(S_{rqi})_{i \leq N_e}$, $q = 1, \ldots, 4$ [CORE/cpp/v3_engine.h:355-361]; the v3.1 accumulator is $H_r(h) = (4N_e)^{-1}\sum_{q,i} h(S_{rqi})$, which for $h = 1\{j \in S\}$ is exactly the coded `pip_plain[j]`.

Premises (named). EX-A; EX-S/H-U (iid uniforms within stream $9 + r$); START (`draw_pi0` exact, $\bar{Z}_0$ exact; N9 PROVED (candidate)); KERNEL (the coded steps realise the T12 kernels; N9 PROVED (candidate), code-checked, CHK_INHERITED VERIFIED WITH AMENDMENT); CAL; SI (stream $9 + r$ independent of streams 0–9; for $r > 4$ implied by IND-T, whose extension audit is pending); MEAS; the potential (uninterrupted) run; worker equivalence for v3.1's concurrent workers [W1/W1-N1/STATEMENT.md:68-91].

Let $\mathcal{A} \in \sigma(\mathcal{C})$ be the successful-calibration event and $E_r = \mathcal{A} \cap \{r \le R'\}$. An identity $\mathbb{E}[Y \mid \mathcal{C}] = b$ "almost surely on E_r" means $\mathbb{E}[1_{E_r} Y \mid \mathcal{C}] = 1_{E_r} b$ almost surely; no island record is asserted outside E_r [W1/W1-N1/STATEMENT.md:94-99].

Lemma 4.1 (N1). *For every island r, every $q \in \{1, \dots, 4\}$ and every jointly measurable $h(S; \mathcal{C})$ with a deterministic bound $|h| \le B_h$, fixed before stream 10, almost surely on $\mathcal{A} \cap \{r \le R'\}$ (localised form):*

$$\mathbb{E}\Big[\hat{Z}_r \cdot \tfrac{1}{N_e} \textstyle\sum_i h(S_{rqi}) \;\Big|\; \mathcal{C}\Big] = Z\,\pi(h), \qquad \textit{hence} \qquad \mathbb{E}[\hat{Z}_r H_r(h) \mid \mathcal{C}] = Z\,\pi(h).$$

No regime assumption on X is made.

Corollary 4.2 (E = M1(e), constant δ_1). *Under N1's named premises, almost surely on E_r in the same localised sense: for $h \in [0,1]$ (as in Lemma N1, defined on $\Omega \cup \{S_0\}$) and $\delta_1 := 1 - \pi(S_0)$: $\mathbb{E}[W_r |H_r(h) - h(S_0)| \mid \mathcal{C}] \le \mathbb{E}[W_r H_r(1\{S \ne S_0\}) \mid \mathcal{C}] = \delta_1 \le 2\delta_1$, attained by $h = 1\{S \ne S_0\}$.*

Status and evidence

W1-N1: PROVED-CHECKED; N1 PROVED under the named premises — cross-family checked, council-passed (round 1); Corollary E PROVED under the same premises; council Codex 9.3 / Opus 8.9 [W1/W1-N1/REPORT.md:44-46]. Statement [W1/W1-N1/STATEMENT.md:95-111]. Stronger than the card in three places (any bounded h; each measured round; constant δ_1). **Not Lean-certified** (Lean not supplied; HQ to waive or open a registrar task). The SoP card still reads N1: PROOF SKETCH [SoP/04_theory.tex:123]; the registry update is HQ's.

4.2 What holds for the reported PIPs

The paper must distinguish four PIP objects [W1C/C-PACK/lemmas/I4_SL_corollaries.md:70-88]:

1. **Exact PIPs** $\mathrm{PIP}_j = \pi(j \in S)$: consistent with rate (Cor. 3.3).
2. **Plain measured-endpoint PIP, islandwise** (shipped $R' = 4$): Corollary 4.2 with $h = 1\{j \in S\}$, evidence-weighted, in expectation given $\mathcal{C}$, for the potential run. This is *not* the study's `pip_primary` and not a statement in probability about the pooled estimate.
3. **Pooled reported PIPs.** The cited cards contain **no in-probability sure-screening theorem for the pooled shipped-$R' = 4$ estimator**. M1(i), status (h), (i): PROVED (by this seat, unchecked) [SoP/04_theory.tex:130], gives $\hat{\mu}(h) \to h(S_0)$ in probability jointly with output for a *fixed finite* declared list under T03♯'s hypotheses, $\mathrm{osc}\, d \le D_* = O_P(1)$ and the stacking configuration $R'_n = 4\lceil \log_2(n+p+1) \rceil$; simultaneous control of $\le (n+p)^2$ declared summaries needs the card's separate configuration $R'_n = 4\lceil (n+p)^6/4 \rceil$ [SoP/04_theory.tex:111]. The finite-R' bound M1(f) is "vacuous at $R' = 4$" [SoP/04_theory.tex:117]. With N1 now proved (Lemma 4.1), the premise "E[W_r $\hat{h}_r$] = π(h)" of PC1c3 used by M1(f),(g),(i) is discharged islandwise; those clauses keep their own statuses and remaining premises (IND-T extension for $R' > 4$, GUARD-W, stacking, T03♯) [W1/W1-N1/REPORT.md:48].
4. **Store-restricted PIPs** $\mathrm{PIP}^{\mathcal{T}}$ (π conditioned on the reported store $\mathcal{T}$), Corollary SL2 below.

Corollary 4.3 (SL2: store-restricted PIPs, jointly with output). *Assume M4(b) (if $S_0 \in \mathcal{T}$: $|\mathrm{PIP}^{\mathcal{T}}_j - \mathrm{PIP}_j| \le \delta_1$), M4(c) (if $S_0 \in \mathcal{S}^+$ and $\delta_1 < \frac{1}{2}$ then $S_0 \in \mathcal{T}$, under LS0 with $\mathcal{T}_{\mathrm{new}} = \mathrm{top}_c(\mathcal{T}_{\mathrm{old}} \cup \mathcal{S}_{\mathrm{added}})$), exact score ordering, a positive-capacity store, truth discovery with output $\mathbb{P}(\mathcal{O} \cap \{S_0 \notin \mathcal{S}^+\}) \to 0$ (PXN or GEN), and $\delta_n = O_P(r_n)$. Then $\mathbb{P}(\mathcal{O} \cap \{\max_j |\mathrm{PIP}^{\mathcal{T}}_j - 1\{j \in S_0\}| > 2\delta_n\}) \to 0$, and the same with any deterministic threshold t_n, $r_n = o(t_n)$.*

Status and evidence

SL2: PROVED (by this seat, unchecked), R34 re-derived; C-PACK council passed. Inputs verbatim: M4(a) MZ-P3: "PROVED (P3 top store, deterministic); P3′ conditional on T03 + discovery"; "candidate: not reviewed, not adopted"; AUTHOR-ONLY; M4(b) same as (a); W1PC1 §3 B.D3(b) COUNTEREXAMPLE repaired with $\|h\|_\infty$; M4(c) N13: PROVED (PC1 claim 5) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:112-114]. Statement [W1C/C-PACK/lemmas/I4_SL_corollaries.md:90-103].

Not claimed / not proved — sampler level

- No in-probability sure-screening or selection theorem for the shipped pooled $R' = 4$ PIPs or MPM; no TV/KL bound for the final pooled estimate under masking (CJ02).
- N1 and Corollary E are about the potential run; delivered records equal potential ones on $\mathcal{O}$ only through GUARD-W (audit open). Worker equivalence, the IND-T extension ($R' > 4$) and START/KERNEL (N9, candidate) remain premises.

5 Reported-HPM selection consistency

5.1 The route: M5(0), operational closure, LS0

By M5(0) (§1) it suffices to show $\mathbb{E}_0[1 - \pi(S_0)] \to 0$ (layer (i), §3) and $D_n \to 0$. LS0 reads no uniforms, so the models it scores are a deterministic function of the data; under the named premises the **operational closure** [OP] gives

$$\bar{\mathcal{O}} \cap \{S_0 \notin \bar{S}^+\} \subseteq \{\text{LS0 from } \tilde{S}_0 \text{ does not score } S_0\}, \qquad \text{so} \qquad D_n \le \mathbb{P}(\text{LS0 misses } S_0)$$

(PROVED (by this seat, unchecked) inside Theorem GEN PROVED-CHECKED; XF VERIFIED as a theorem under (OP) in W1-PXN) [W1/W1-GEN/STATEMENT.md:169-171]; [W1/W1-PXN/REPORT.md:43]. Every theorem in this section is therefore a statement about two data events: the start mode lies in a region from which the greedy search reaches S_0, and the search does reach it within its move cap. **None of them is a delivery claim**; the liminf form $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$ is algebra from $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$.

5.2 The start-mode centre: e4 is not needed (C-PACK item 1)

Lemma 5.1 (UM). *Under W1-N3 (H1)–(H4) (correct encoding of the `Start` tables, exact arithmetic, enumeration cap ≥ 1, C++ execution limits), if* $\arg\max_\Omega f = \{R\}$ *then* `start_mode(st, cap).S` $= R$ *(from N3(c) with $N = 1$, or N3(a)); the truncation flag is irrelevant. In particular this holds on* $\{\pi_0(R) > \frac{1}{2}\}$.

The positive consistency proofs of PXN, GEN and M5(b1-LH) use the LS0 centre only to identify it with a deterministic reference $R \in \{S^\circ, S^*\}$ on $\{\pi_0(R) > \frac{1}{2}\}$; hence premise e4 is replaced by **e4′**: (i) the LS0 centre is `start_mode(st, cap).S` of the coded routine, or a replacement returning the unique maximiser whenever it is unique (C-PACK MR1); (ii) the routine uses the same retained indices, K, ordered actual partition, full-response tables and prior as the mathematical π_0, and the coded centre is a deterministic measurable data function on every branch; (iii) wherever LS0 is required its emitted centre is this function, with OP-LS0/scored-set/guard contracts unchanged; (iv) the consumer's EX-A [W1C/C-PACK/lemmas/I1_e4_replacement.md:141-160]. The lexicographic tie rule and the truncation fallback are needed by no positive consistency statement; Theorem GROW needs only "some maximiser" of γ_0 [W1C/C-GROW/lemmas/00_MAIN.md:31-33]. A binary64 version (Lemma UM-FP, under FP-TAB $\Delta(R) > 2(\eta+\varepsilon)$) covers the start-mode identification step only; FP-TAB is untested for the coded scorer.

Status and evidence

C-PACK items 1–4: PROVED-CHECKED (council round 1: Codex gpt-6.1-sol xhigh 9.0, Claude Opus 5.5 high 8.7, all six gates PASS) [W1C/C-PACK/REPORT.md:19-21]; item 1 PROVED; XF-checked (R33 GAP → 7 fixes), re-read R38/R39, council. W1-N3 (the code's actual start-mode behaviour; counterexample above the cap 10^6; reference algorithm TDP): PROVED-CHECKED (council Sol 9.3 / Opus 8.6); "e4 for integrated LS0 stays OPEN" as an integration item [W1/W1-N3/REPORT.md:12-14].

5.3 Theorem PXN: masked correlated design, p ≫ n (LH-iid)

Family LH-iid. Fixed $g \geq 2$, $c \geq 1$, $v \geq 0$, $\rho \in (0,1)$, links $0 < |\psi_i| < \rho$, nonzero $\beta_i, \beta_h^{\mathrm{vis}}$, $\sigma^2 > 0$, $A \geq 1$. $p_n = G_n g$, $18 \leq p_n \leq n^A$, $p_n \to \infty$. Rows iid $N(0,\Sigma)$: groups of size g equicorrelated at ρ; c linked group pairs with rank-one cross-correlation through the hubs ($\mathrm{corr}(j_a, j_b) = \psi$); v visible groups; the rest null. Truth: $\theta_{j_{a_i}} = \beta_i$, $\theta_{j_{b_i}} = -\psi_i \beta_i$, visible hubs β_h^{vis}; $s_0 = 2c + v$. The b-hubs are *masked* (zero population covariance of their block with the signal); reference $S^\circ = S_0 \setminus \mathcal{M}$. **(SIZE)** $g \leq B_n < 2g$ eventually ($9 \leq g \leq 16$ asymptotically) [W1C/C-PACK/item5_PXN_consolidated/00_statement_and_conventions.md:8-30]. Method: v3 + LS0 (one production local search from the exact start mode, centre emitted, cap ℓ_n), under (OP) = EX-A, EX-S, CAL, SR, GUARD, OP-LS0, with the **exact-branch convention** (blocks, tables, π_0, $\tilde{S}_0$ and "LS0 of $\bar{R}^{+}$" are the deterministic data functions on every branch).

Write $s = |S^\circ| = c + v$, $\bar{\psi} = \max_i |\psi_i|$, $q_R = \sqrt{(1-\bar{\psi}^2)/(1-\rho^2)}$, $t^* = (q_R - 1)/(q_R\rho - \bar{\psi}) \in (\bar{\psi}, \rho)$ and $I(t;r) = -\frac{1}{2}\log[(1-r^2)(1-t^2)/(1-rt)^2]$.

Theorem 5.2 (PXN). *Under LH-iid with (SIZE), $18 \leq p_n \leq n^A$, $p_n \to \infty$, s_0 fixed, and (OP), as $n \to \infty$:*

(R) $\mathbb{P}(\text{blocks} \neq \text{groups}) \to 0$, *summable, and for every fixed $C > 0$, $\leq p^{-C}$ for $n \geq \max\{n_1, n_0(A, C, \rho, \bar{\psi})\}$ (n_1 the SIZE onset); explicitly $\leq \frac{2}{\sqrt{\pi}(1-\rho)} p^2 \sqrt{n}\, e^{-(n-4)I^*}$ for $n \geq 5$ with SIZE at n, $I^* = I(t^*; \bar{\psi}) > 0$.*

(S) $1 - \pi_0(S^\circ) = O_P(\sqrt{\log n / n})$; $\pi_0(S_0) = p^{-c} n^{-c/2} e^{O_P(1)} \to 0$, *more precisely* $\log \pi_0(S_0) + c \log p + \frac{c}{2}\log n \to_d \sum_{i \leq c} \log \frac{s+i}{2} + \frac{1}{2}\chi^2_c$ *(the masked regime).*

(E) $\mathbb{P}(\tilde{S}_0 = S^\circ) \to 1$.

(T) $\mathbb{P}(\text{LS0 scores } S_0) \to 1$ *and* $\mathbb{P}(\text{LS0 returns } S_0 \text{ with } \texttt{local_certified}) \to 1$.

(P) $1 - \pi(S_0) = O_P(\sqrt{\log n / n})$ *under the joint law of* (X^r, ε).

(C) For every n: $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2\mathbb{E}_0[1 - \pi(S_0)] + D_n$, $D_n \leq \mathbb{P}(\text{LS0 misses } S_0) \to 0$; *hence* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$ *and* $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$.

(K) LS0 costs $\leq 1 + (\ell_n + 1)\max_{k \leq K}[p + k(p-k)] = O(p^2 \log(n+p))$ scores; on the joint event $\mathcal{T}_n$ (lattice arrival, (L), $\ell_n \geq c$) it uses $c + 1$ scans.

Corrected centring remark. The finite-n centring added after the W1-PXN council claimed a remainder $O_P(\sqrt{\log n/n})$; this fails for slowly growing p. The correct remainder is $O_P(\sqrt{\log n/n + 1/p})$ (prior correction $d_p = \sum_{i \leq c} \log[2p/(2p - s - i)] = O(1/p)$); the limit law in (S) and the headline are unaffected, and no rate of convergence to the χ^2 law is asserted [W1C/C-PACK/item5_PXN_consolidated/03_start.md:111].

Status and evidence

W1-PXN: PROVED-CHECKED (council round 1: Codex 8.6, Opus 8.3, all gates PASS) [W1/W1-PXN/REPORT.md:3-6]. Theorem as stated in the consolidated supplement [W1C/C-PACK/item5_PXN_consolidated/00_statement_and_conventions.md:97-119]; supplement fidelity FAITHFUL-WITH-FIXES (R36), all fixes applied. C-PACK addenda (separately identified, not silently merged): e4 → e4′ and Lemma T(i)'s off-event centre identity replaced by $\mathbb{P}(C_0 \neq S^\circ) \leq \mathbb{P}(\mathcal{A}_n^c) + \mathbb{P}(\tilde{\pi}_0(S^\circ) \leq \frac{1}{2})$. **A-dependence**: no step

uses $p \le n$; A enters only constants and onsets; a sufficient onset envelope is $n/(\log n)^3 \ge C(A, \text{fixed parameters})$ plus SIZE; no uniformity over growing A.

5.4 Theorem GEN: a general correlated class with a bounded relevance set

Fixed-design model G, standardised design, $s_0 \ge 1$ fixed, $p \le n^A$; method v3.1 + LS0 as specified with cap ℓ_n; premises EX-A, EX-S, CAL, SR, GUARD-W, OP-LS0, e4 (read e4′), complete scans; IND-T and the c_4 search when $R' > 4$. Notation: $\Lambda_T = \|(I - P_T)\mu\|^2$, $\Delta_T = \Lambda_T/(n-1)$, $v_T = \sigma^2 + \Delta_T$, $\delta_{T,l} = u_{T,l}^\top \mu$ with $u_{T,l}$ the normalised residual of x_l on X_T.

- **(H0)** the T03♯ hypotheses with BM-∞.
- **(H1) bounded separated relevance set**: deterministic $\mathcal{V}_n \supseteq S_0$, $|\mathcal{V}_n| \le \bar{v}$: (H1a) $\lambda_{\min}(\hat{R}_{\mathcal{V}\cup\{l\}}) \ge \phi_0^2$ for every l; (H1b) SEP(c_S): $\max\{\Lambda_S - \Lambda_{S\cup\{l\}} : S \subseteq \mathcal{V}, l \notin \mathcal{V}\} \le c_S(n-1)\phi_0^4\theta_{\min}^{*2}$ eventually, fixed $c_S \in [0,1)$.
- **(H2) centre certificate on the actual blocks**: deterministic $S^* \subseteq \mathcal{V}$, $|S^*| \le \bar{v}$, $U_b^* = S^* \cap C_b$: (H2a) within-block SE: $\lambda_{\min}(\hat{R}_T) \ge \phi_0^2$ for every nonempty $T \subseteq C_b$, every b; (H2b) VIS: $\|(P_{U\cup U_b^*} - P_U)\mu\|^2 \ge M_n v_U \log n$ for every b and every $U \subseteq C_b$ with $U_b^* \not\subseteq U$, for a deterministic $M_n \to \infty$; (H2c) NL: with the leakage family $\mathcal{L}(S^*) = \{(b,T,l) : l \in C_b \setminus U_b^*,\ U_b^* \subseteq T \subseteq C_b \setminus \{l\}\}$ and fixed $C_L < \infty$, $\kappa > 0$, $\varepsilon_0 > 0$: $\delta_{T,l}^2 \le C_L\sigma^2 \log n$ on $\mathcal{L}(S^*)$ and, if $p > s^*$, eventually $(p-s^*)^{-1}\sum_{l\notin S^*} \max_{T:(b(l),T,l)\in\mathcal{L}(S^*)} \exp(E_{T,l}) \le n^{1/2-\kappa}$, where $h_n = \sqrt{2(A+\varepsilon_0)\log n}$ and $E_{T,l} = \min\{\delta_{T,l}^2/(2\Delta_T),\ \delta_{T,l}^2/(2v_T) + |\delta_{T,l}|\sigma h_n/v_T\}$, with $E_{T,l} = 0$ when $\delta_{T,l} = 0$ (in particular when $\Delta_T = 0$); the average is omitted when $p = s^*$ ($\mathrm{NL}_{\mathrm{avg}}(\kappa)$). Convention for (H1b): the maximum is 0 if $\mathcal{V} = [p]$ [W1/W1-GEN/STATEMENT.md:56-75].

Canonical certificate (C1′). Under (H2a) and $\|\mu\|^2 \le Mn$, let γ_b be the coefficients of the projection of μ on the block's own columns. If $\gamma_b = \gamma_b^L + \gamma_b^S$ (disjoint supports) with $\|X_{C_b}\gamma_b^S\|^2 \le \eta\sigma^2 \log n$ and $\sqrt{(n-1)\phi_0^2}\min|\gamma_{b,j}^L| \ge \sqrt{\eta\sigma^2\log n} + \sqrt{M_n(\sigma^2+2M)\log n}$ whenever $\operatorname{supp}\gamma_b^L \ne \emptyset$, for a fixed $\eta \in [0, (\sqrt{2A+1} - \sqrt{2A})^2)$ and some $M_n \to \infty$, then $S^* := \bigcup_b \operatorname{supp}\gamma_b^L$ satisfies (H2b) and $\mathrm{NL}_\eta(S^*)$ [W1/W1-GEN/STATEMENT.md:89-97]; **amendment P-L8.1**: (H2c) is then supplied with $C_L = \eta$ and $\varepsilon_0 > 0$ small enough that $\eta/2 + \sqrt{2\eta(A+\varepsilon_0)} < \frac{1}{2}$, and any fixed $0 < \kappa \le \frac{1}{2} - \eta/2 - \sqrt{2\eta(A+\varepsilon_0)}$ ($\eta = 0$: any $\varepsilon_0 > 0$, $0 < \kappa \le \frac{1}{2}$) [W1C/C-PACK/patches/P_L8_certificate.md:7-13]. Masked truths are truths with zero (or negligible) block coefficient; they are recovered by the descent. These are oracle conditions in $(X, \beta, \sigma^2, \text{partition})$, not an observable test.

Theorem 5.3 (GEN). *(C1) Under G, $\|\mu\|^2 \le Mn$, $p \le n^A$, (H2), $s^* \le \bar{v}$: $1 - \pi_0(S^*) \to 0$ in probability and $\mathbb{P}(\tilde{S}_0 = S^*) \to 1$. (C2) Under G, R, BM-∞, (H1): on events of probability $\to 1$, simultaneously for every centre $C \subseteq \mathcal{V}$, the uncapped greedy path keeps its centres in $2^{\mathcal{V}}$, reaches S_0 in $\tau(C) \le 2^{\bar{v}} - 1$ moves, and S_0 has no strictly improving feasible neighbour. (C3) Under (H0), (H1), (H2) and the premises, for $n \ge n_0$ with $\ell_n \ge 2^{\bar{v}} - 2$: $D_n \le \mathbb{P}(\tilde{S}_0 \ne S^*) + O(1/\log n) \to 0$, hence $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \to 0$ and $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$. (C3-fav) If $S^* = S_0$, (H1) is not needed. (C4) Random designs, X^r independent of ε, β possibly X-measurable: if every deterministic sequence in X-events $\mathcal{A}_n$ with $\mathbb{P}(\mathcal{A}_n) \to 1$ satisfies (H0)–(H2) with the* same *constants and envelopes, the conclusions hold under the joint law.*

Instances (C5). (a) LH with exact Gram, SIZE: $S^* = S_0 \setminus \mathcal{M}$, $\mathcal{V}$ = signal units (input: blocks = groups, W1-EXP Lemma 4.2 via Thm5-LH, SAME-FAMILY-ONLY). (b) **LH-iid, now without candidate inputs**: partition recovery and the correlation bound come from W1-PXN's repaired Theorem R (or route R′) and Lemma CR/[CONC′] ($\delta_n = 32\sqrt{2(A+1)\log n/(n-1)}$, operator error $O((\log n)^{3/2}n^{-1/2})$), with the exact-branch convention; GEN's other hypotheses, envelopes and premises are retained (C-PACK Lemma I2; "unconditional" means removal of the two candidate inputs only; onset $\approx 10^{10}$ for typical constants) [W1C/C-PACK/patches/P_GEN_LHiid.md:18-24]; [W1C/C-PACK/lemmas/I2_LHiid_unconditional.md:

80]. (c) W (exact Gram): favourable, $S^* = S_0$. (d) **Study generator**, asymptotic sequence: groups of 5, $p_n = 5G_n \to \infty$, $p_n \le n^A$, $s_0 = 14$ in one of the study's two high-p layouts, a fixed scale $c > 0$ (amendment P-SEL.2), iid $N(0, \Sigma_s)$ rows: **S1, S3, S4, S6 covered** by (C3-fav) via (C4) (Lemma SEL); S2 and S5 outside the class. (C6) The bounded-size conditions cannot be dropped from the LS0 route: an explicit design satisfying every hypothesis except $|\mathcal{V}_n|, s^* \le \bar{v}$ has $\mathbb{P}(\text{LS0 misses } S_0) \to 1$ while $\pi(S_0) \to 1$ — a counterexample for the *LS0 route with cap* ℓ_n, not for the full method.

Status and evidence

W1-GEN: PROVED-CHECKED (council round 1: Sol 9.0, Opus 8.9, all six gates PASS) [W1/W1-GEN/REPORT.md:17-25]; statement v2.3 [W1/W1-GEN/STATEMENT.md:84-154]. C-PACK item 2 (LH-iid without candidate inputs) PROVED; XF-checked (R35 V-W-A, 3 fixes), R38/R39, council; item 3: Lemma SEL and L8 (C1′) each VERIFIED-WITH-AMENDMENTS by a dedicated xhigh check (R31, R32), amendments re-checked (R37); "No hypothesis or conclusion of Theorem GEN changes" [W1C/C-PACK/patches/P_SEL_L6d.md:42-44]. Finite n: at the study's size ($n = 200, p = 500$) the visibility margins are 0.16–0.94 (< 1); no finite-n claim.

5.5 Theorem GROW: growing relevance sets, including study S2

Same setting as GEN, but LS0 runs with the **method requirement MR1**: move cap $\ell^{\text{LS0}} := \ell_n + |\tilde{S}_0|$ (all other LS0 rules unchanged; the final local search keeps ℓ_n) [W1C/C-GROW/METHOD_REQUIREMENTS.md:1-6]. For a deterministic null set $\mathcal{P}_n$ let $Q = S_0 \cup \mathcal{P}_n$ and $\mathcal{L} = \{T : S_0 \subseteq T \subseteq Q\}$.

- **(H0)** the T03♯ hypotheses with BM-∞.
- **(G1)** $|Q| \le C_Q \log n$ (C_Q fixed), $|Q| \le K$, and either $Q = [p]$ or $K \ge 2|Q| + 32s_0 + 1$, for $n \ge n_0$.
- **(G2)** $\lambda_{\min}(\hat{R}_{Q\cup\{l\}}) \ge \phi_0^2$ for every $l \in [p]$.
- **(G3)** on the actual blocks: (G3a) within-block SE: $\lambda_{\min}(\hat{R}_T) \ge \phi_0^2$ for every nonempty $T \subseteq C_b$, every b; (G3b) TV: for a deterministic $M_n \to \infty$, every $j \in S_0$ and every $U \subseteq C_{b(j)} \setminus \{j\}$: $\delta_{U,j}^2 \ge M_n \log n$; (G3c) NEG($\eta$) for a fixed $\eta \in [0, (\sqrt{2A+1} - \sqrt{2A})^2)$: every $l \notin Q$ and U with $S_0 \cap C_{b(l)} \subseteq U \subseteq C_{b(l)} \setminus \{l\}$: $\delta_{U,l}^2 \le \eta\sigma^2 \log n$.
- **(G4)** (G4-χ) $|\mathcal{P}_n| \le (1 - \delta_0) \log n$ for a fixed $\delta_0 \in (0, 1)$; or (G4-dir) the add directions $\{u_{T,l} : T \in \mathcal{L}, l \notin T\}$ number at most pN_n with $\log N_n = o(\log n)$.

Theorem 5.4 (GROW). *Let* $\text{NA}(\mathcal{L}) = \{F(T \cup \{l\}) < F(T)$ *for all* $T \in \mathcal{L}$ *with* $|T| < K$ *and all* $l \notin T\}$ *and* $\text{TL}(\mathcal{L}) = \{F(A) < F(T)$ *for every* $T \in \mathcal{L}$ *and every feasible neighbour* A *of* T *with* $S_0 \not\subseteq A\}$ *(strict inequalities). (C1) On* $\text{NA}(\mathcal{L}) \cap \text{TL}(\mathcal{L})$*, from every centre* $C \in \mathcal{L}$ *the uncapped §6.10 path deletes one element of* $C \setminus S_0$ *per move and stops at* S_0 *after* exactly $|C \setminus S_0|$ *moves; the capped search scores* S_0 *if* $\ell \ge |C \setminus S_0| - 1$ *and ends certified at* S_0 *without* `local_limit` *if* $\ell \ge |C \setminus S_0|$*. (C2) Under G,* $\|\mu\|^2 \le Mn$*,* $p \le n^A$*,* $|Q| \le C_Q \log n$*,* $|Q| \le K$*, (G2) and BM-∞:* $\mathbb{P}(\text{TL}(\mathcal{L})) \to 1$*. Under the same hypotheses without BM-∞, and with (G4):* $\mathbb{P}(\text{NA}(\mathcal{L})) \to 1$*. (C3) Under G,* $\|\mu\|^2 \le Mn$*,* $p \le n^A$*, (G1) and (G3):* $\mathbb{P}(S_0 \subseteq \tilde{S} \subseteq Q$ *for every maximiser* $\tilde{S}$ *of* γ_0 *on* $\Omega) \to 1$*. (C4) Under (H0), (G1)–(G4), the premises and MR1:* $D_n \le \mathbb{P}($*sandwich fails*$) + \mathbb{P}(\text{NA}^c) + \mathbb{P}(\text{TL}^c) \to 0$*; hence* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \to 0$ *and* $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$*. (C4)(b) Under (G4-χ) the production cap* ℓ_n *suffices.*

Instances (C5). (a) *AR(1), exact Gram*: $X^\top X = (n-1)\Sigma$, $\Sigma_{ij} = \rho^{|i-j|}$, fixed $\rho \in (-1, 1) \setminus \{0\}$, any $p \le n - 1$ (fixed or growing), fixed s_0, fixed nonzero θ^*, fixed $\sigma^2 > 0$, any block partition with blocks of size ≤ 16, separation $\sum_{k \in S_0 \setminus \{j\}} |\theta_k^*||\rho|^{|j-k|} \le \frac{1}{2}|\theta_j^*| \frac{1-\rho^2}{1+\rho^2}$ for every $j \in S_0$. Set $d_n = \lceil \log n / (2 \log(1/|\rho|)) \rceil$, $\mathcal{P}_n^0 = \{l \notin S_0 : \min_{k \in S_0} |l - k| \le d_n\}$, $Q_n^0 = S_0 \cup \mathcal{P}_n^0$; use $\mathcal{P}_n = \mathcal{P}_n^0$ if $Q_n^0 = [p]$ or $p \ge 2|Q_n^0| + 32s_0 + 1$, and $\mathcal{P}_n = [p] \setminus S_0$ otherwise. Then (H0), (G1)–(G3), (G4-dir) hold and (C4) applies (with MR1); $|Q| \le 4s_0 d_n + 34s_0 + 1 = O(\log n)$ (an envelope: it is not claimed that the start mode has unboundedly many extras). (c) *AR(1) with iid Gaussian rows = study S2*: raw rows iid $N(0, \Sigma)$, $\Sigma_{ij} = \rho^{|i-j|}$, fixed $\rho \in (-1, 1) \setminus \{0\}$, columns centred and standardised, $p \le n^A$ (any fixed $A \ge 1$; $p \gg n$ allowed); on

standardised columns $Y = X\theta^* + C\varepsilon$ with fixed nonzero θ^*, $\varepsilon \sim N(0, \sigma^2 I_n)$ independent of the raw design, $\sigma^2 > 0$ fixed, and $y = Y/(\|Y\|/\sqrt{n-1})$ inside the score; separation and the piecewise relevance set as in (a); any X-measurable partition with blocks of size ≤ 16; implementation premises and MR1 retained. Under the joint law of (X, ε) the three events (sandwich, NA, TL) are proved directly; the generator's S2 layouts ($p = 500, 1000$; 14 truths; $\rho = 0.5$) satisfy separation for every scale $c > 0$, so the result covers asymptotic sequences extending them — *not* the generator's fixed $n = 200$. (b) *S5 population covariance at fixed* $p = 100$ (diagnostic layout): an illustration only — for fixed p the production run eventually enumerates all models ($n \gtrsim 2.5 \cdot 10^{27}$ for $p = 100$, before the (G4-χ) onset), so T03♯ alone gives the conclusion there; Theorem GROW adds content beyond T03♯ + enumeration only for $p \gg \log n$ [W1C/C-GROW/STATEMENT.md:118-149].

Status and evidence

C-GROW: PROVED-CHECKED (council round 1: Sol 9.2, Opus 9.0, all six gates PASS); L6 (random AR(1)) by a parallel prover (Codex Sol), checked by R07 Sol xhigh VERIFIED and R06 Opus VERIFIED-WITH-AMENDMENTS [W1C/C-GROW/REPORT.md:24-30]; statement v2.5 [W1C/C-GROW/STATEMENT.md:84-117]. Inputs: M5(0) PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5)); T03♯ as above; [OP] PROVED (by this seat, unchecked) inside GEN. Premise list keeps e4 verbatim, but the proof uses only "the emitted LS0 centre is some maximiser of γ_0". Finite n: the asymptotic events fail at tested n; S5 needs $\log n \gtrsim 96$.

5.6 Reported HPM under misspecification (C-MIS Corollaries H, D, D′)

Corollary 5.5 (H). *At every n, for any candidate design and data law on valid inputs and any feasible T (deterministic or design-measurable), with* $D_n(T) := \mathbb{P}(\bar{\mathcal{O}} \cap \{T \notin \bar{S}^+\})$*:* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq T\}) \leq 2\mathbb{E}[1 - \pi_{\mathcal{C}}(T \mid y)] + D_n(T)$.

Corollary 5.6 (D, D′). *Under GR with a fixed candidate design, method v3.1 + LS0 with GEN's premises, and GEN's (H0)–(H2) (or (H0), (H2) with* $S^* = S_0^{\mathcal{C}}$*) for the candidate problem with* $\sigma^2 := \sigma_{\mathcal{C}}^2$ everywhere *in GEN and* $\mu := \mu^{\mathcal{C}}$*, on the actual candidate-only partition:* $D_n^{\mathcal{C}} \to 0$, $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0^{\mathcal{C}}\}) \to 0$, $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0^{\mathcal{C}}\}) = 1 - \limsup \mathbb{P}(U_n)$*; random candidate designs via GEN (C4).* ***D′*** *(instance on GEN's W family): the omitted variable is a noisy copy of a candidate* j_2 *that is null in the full model, and the reported HPM contains the proxy* j_2 *in place of the omitted k.*

Status: C-MIS PROVED-CHECKED; Cor. D inherits GEN's premises (LS0 as specified, e4, GUARD-W/OP-LS0, BM-∞); no delivery claim [W1C/C-MIS/STATEMENT.md:109-120].

5.7 The Song–Liang-style general-design result and study coverage

Reading for the paper. GEN and GROW together are the reported-model analog of Song–Liang's MAP consistency on general designs: Song–Liang's identifiability Condition 3 and screening Condition 5 are replaced by (i) Gram floors on a relevance set, (ii) T03♯'s anchored sparse eigenvalues and beta-min, and (iii) block-level visibility of the truths and negligibility of the outside variables on the method's own partition — bounded relevance sets with masking allowed (GEN, certificate (C1′)), or relevance sets of size $O(\log n)$ without masked truths at block level (GROW). Both are asymptotic and joint with output.

Scenario	design (generator `Scripts/Data_generators/gen_dev_data.R`)	covering result (asymptotic sequence, not $n = 200$)	status
S1, S3, S4, S6	groups of 5, iid rows, 14-signal layouts	GEN (C5)(d), favourable case via (C4), Lemma SEL (amended P-SEL.1–3)	GEN PROVED-CHECKED; SEL VERIFIED-WITH-AMENDMENTS
S2	AR(1), $\rho = 0.5$, iid rows	GROW (C5)(c), $p \le n^A$ incl. $p \gg n$, under MR1	C-GROW PROVED-CHECKED
S5	compound symmetry (within 0.7, between 0.3)	**frontier**: only GROW (C5)(b), fixed $p = 100$ population-covariance illustration; random-design S5 and factor designs with growing p not proved	—

Sources: [W1/W1-GEN/STATEMENT.md:135-143]; [W1C/C-GROW/STATEMENT.md:150-156]; [W1C/C-GROW/REPORT.md:12-15].

6 Delivery

All results of §5 are joint with output: they leave $\mathbb{P}(U_n)$ free. C-DLV asked whether $\mathbb{P}(U_n) \to 0$ for v3.1 + LS0 under a declared polynomial resource allocation. **For the shipped calibration this target remains open**; what is proved is an exact reduction to one calibration event, and — with a one-line calibration fallback that HQ has adopted on branch `w1c/ls0-merge` behind a switch — $\mathbb{P}$(no output) $= 0$ (Proposition 6.3).

Setting. Method $\mathfrak{M}$ = v3.1 + LS0 as specified, $R' = 4k$ declared (k fixed); premises EX-A (also as a unit-cost model for exact reals, including exp and log), EX-S, SR, OP-LS0, e4, GUARD-W, IDEAL-W, MACH, CONFIG, TDP-IMPL; valid declared seed and allocation; **CAL is not assumed**. Classes: LH (exact Gram), LH-iid, the W1-GEN class; fixed s_0, $p \le n^A$. $\mathcal{U}_n$ = no output; $\mathcal{SL}_n$ = status `schedule_limit` [W1C/C-DLV/STATEMENT.md:10-46].

6.1 The exact reduction

Theorem 6.1 (D; C-DLV). *For every $n \ge 3$: (i) $\mathcal{U}_n \subseteq \mathcal{SL}_n \cup \mathcal{N}_n$ with $\mathbb{P}(\mathcal{N}_n) = 0$, and $\mathcal{SL}_n \subseteq \mathcal{U}_n$; (ii) $\mathcal{SL}_n = \{$calibration escalates to tier 1, and some tier-1 grid construction fails$\}$; (iii) a construction fails iff (α) its interval count exceeds the tier cap T_e (pilot stages plus appended bisection points), or (β) non_progress (requires a d-range on the candidate set above $2\sqrt{c_0}\,2^{60} \approx 7.5 \cdot 10^{17}$, $c_0 = \log(10/9)$), or (γ) failed validation at a point snapped to 1. Hence $\mathbb{P}(\mathcal{U}_n) \to 0 \iff \mathbb{P}(\mathcal{SL}_n) \to 0$.*

Supporting results: **(P1)** inventory of every exit of `v3_fit` without estimates (AUDITED by code reading); **(P2) Lemma R**: operations, words, preflight bytes and preflight seconds are bounded by explicit polynomials in (n, p_{raw}) for every valid input, seed and uniform realisation, so under MACH and the declared allocation no resource path fires; **(P3) Lemma S**: on the classes the input/degenerate statuses have probability 0 and under EX-A no scorer call fails; **(P5)** grid lemmas (exact identity $\rho_V(u, v) = q_u(M)q_{2v-u}(M)/q_v(M)^2$; pilot schedules have $\le T_e$ strictly increasing stages ending at 1; dominance or two-model sufficient conditions) [W1C/C-DLV/STATEMENT.md:52-67].

(O) The open event, exact form. $\mathbb{P}(\mathcal{SL}_n) \to 0$ on the classes, i.e. $\mathbb{P}(E_n \cap F_n^1) \to 0$, E_n = some tier-0 trigger fires, F_n^1 = some tier-1 construction fails through (α)–(γ). Sufficient routes: (O-a) $\mathbb{P}(E_n) \to 0$; (O-b) control of (α)–(γ) for the realised tier-1 candidate sets and pilot schedule jointly with the trigger. The pilot stage count is the CJ06 / Conjecture P question; neither a pilot bound alone nor endpoint posterior concentration suffices. No design sequence is known to satisfy the (P5) sufficient conditions; in all 134 production fits run by C-DLV no tier-0 trigger fired, so tier 1 was never reached [W1C/C-DLV/STATEMENT.md:68-84].

Corollary 6.2 (D′: joint consistency with the delivery term). *On LH, LH-iid and the GEN class, under the respective hypotheses of W1-GEN (C1)+(C2) with T03♯, and W1-PXN (T), (P):* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) \geq 1 - o(1) - \mathbb{P}(\mathcal{SL}_n)$, *pathwise from* $\mathcal{O} \cap \{\hat{S}^+ \neq S_0\} \subseteq \{\pi(S_0 \mid y) \leq \frac{1}{2}\} \cup \mathcal{K}_n^c$, $\mathcal{K}_n$ *= the prescribed deterministic LS0 scores* S_0. *CAL is not used; the LH exact-Gram instance carries the input status SAME-FAMILY-ONLY.*

Status and evidence

C-DLV: PARTIAL — the ticket target (T) "$\mathbb{P}$(no output) → 0 … for v3.1 + LS0 under a declared polynomial allocation" remains open for the shipped calibration. The reduction package: council round 2 passed (Codex 9.0, Opus 8.9, all six gates PASS) [W1C/C-DLV/REPORT.md:22-27]. Per clause: P1 AUDITED (code reading); P2 PROVED-CHECKED (council round 2); P3 S.1, S.2 PROVED-CHECKED, S.3 AUDITED (code reading); P4 PROVED-CHECKED (relative to the P1/P3 audits); P5 PROVED-CHECKED; C2 PROVED-CHECKED; O OPEN [W1C/C-DLV/REPORT.md:61-70]. Proofs `W1C/C-DLV/lemmas/L1_resources.md–L5_insertions.md`; Theorem D [W1C/C-DLV/lemmas/L4_delivery.md:13-22].

6.2 Proposition F: the total tier-1 fallback

C-DLV found (as a finding on a shipped component, not made by the ticket) that `schedule_limit` is the only no-output path not excluded by the declared allocation and the idealisation premises, and that freezing a valid fallback grid removes it [W1C/C-DLV/lemmas/L4_delivery.md:79-93]. The HQ red-team audit A1 found HOLDS-WITH-AMENDMENTS and no fatal finding; its one MAJOR finding is that the combination with C-GROW's LS0 cap needs amended resource envelopes and allocation (the old time shape is not justified after the change: at $p \propto n$ the old core term is of order $n^5 \log^3 n$, while the sufficient extra LS0 term is of order n^6) [W1C/HQ_AUDIT/A1_propF_redteam.md:7-19]. The statement below is A1's "Exact corrected statement" [W1C/HQ_AUDIT/A1_propF_redteam.md:60-75].

Proposition 6.3 (F: total tier-1 fallback, with the C-GROW cap; C-DLV + A1). *Consider the specified v3.1 method with deterministic LS0 from the exact e4 start mode, complete scans, emitted centre and scored-set HPM retention. Let* $\ell_n = \lceil \log_2(n + p + 1) \rceil$, $K = \min(p, \lfloor n/2 \rfloor)$, *and set LS0's move cap to* $a_n = \ell_n + |\tilde{S}_0|$; *all other local-search caps remain* ℓ_n. *Keep fixed declared* $R' = 4k$, *fixed worker count, and the other CONFIG restrictions of C-DLV. Assume EX-A with its unit-cost interpretation, IDEAL-W, EX-S, SR, e4, GUARD-W, TDP-IMPL and MACH as stated in C-DLV, and OP-LS0 with the cap replaced by* a_n, *with valid declared seed/allocation and model G on the stated classes.*

Fallback. *At the first failed tier-1 grid construction, terminate that tier and select its last successful construction if one exists, otherwise its completed pilot schedule. Preserve the tier-1 budgets and inherited scored state, freeze the selected complete grid through the normal frozen-record construction before final randomness, and continue with the prescribed final run and LS0. Retain the failure/fallback metadata; make no further calibration attempts and do not promote a partial failed grid to a fallback. Leave other branches unchanged.*

Allocation. *With* $q(p) = p + p^2/4$, $W_G := \bar{W} + C\,K\,q(p)\,[n^3 + p(n+2)]$ *and* $M_G := \bar{M} + C\,K(K+1)$ *($\bar{W}$, $\bar{M}$ the work and memory envelopes of C-DLV Lemma R, C a large enough implementation-dependent constant, raw dimension in the envelopes), choose*

$$\mathrm{mem}_n \geq \max\{\beta M_G(n, p_{\mathrm{raw}}) + \beta_0, P_0, P_1\}, \quad \mathrm{time}_n \geq \max\{2[\tau W_G(n, p_{\mathrm{raw}}) + \tau_0], \bar{P}_t\},$$

`allocation$mem_gb` $\geq \mathrm{mem}_n/2^{30}$, `allocation$hours` $\geq \mathrm{time}_n/3600$, *where* $P_0, P_1, \bar{P}_t$ *are Corollary R's preflight bounds and certificate budgets remain included.*

Then for every admissible $n \geq 3$, $\mathbb{P}(\text{no output}) = 0$. *On LH exact Gram, LH-iid and the W1-GEN class, under the respective statistical hypotheses and inherited dependency qualifications used in Corollary 6.2,* $\mathbb{P}(\hat{S}^+ = S_0) \to 1$ *under the original unconditioned data law. CAL, restricted-ESS success, and any bound on the probability of triggering the fallback are unnecessary. With the original LS0 cap* $a_n = \ell_n$, *the original Lemma R envelopes suffice.*

Status and evidence

C-DLV (F1) Proposition F: PROVED-CHECKED for the patched calibration law (R04; council round 2) [W1C/C-DLV/REPORT.md:70]. HQ audit A1 (fresh Codex Sol xhigh red team): HOLDS-WITH-AMENDMENTS, no FATAL finding; the corrected statement above (total fallback, cap a_n, envelopes W_G, M_G) is A1's and has **not** itself been through a separate council. A1 also independently confirms that the larger-cap LS0 path contains the smaller-cap scored set [W1C/HQ_AUDIT/A1_propF_redteam.md:56].

Remark 6.4 (what Proposition F does and does not say)**.** (1) The fallback changes the calibration law only on the event where the shipped code returns no estimates; final-run validity holds conditionally on any frozen grid (calibration contract). The fallback grid keeps the contract's T_e cap but need not have the candidate grid's restricted-ESS property. (2) Implementation precision (A1): the shipped code sets `out$grid=NULL` on *any* grid failure, so changing only the status line would fail validation; the patch must carry a complete grid through the tier return and the normal freeze, and must not use a failed construction's partial prefix [W1C/HQ_AUDIT/A1_propF_redteam.md:43]. (3) It is a theorem about an abstract unit-cost, unbounded-word machine with MACH's allocation-success and environment clauses; it is *not* a reliability statement about the shipped binary or the default 8 GiB / 24 h allocation. (4) The classes are LH, LH-iid and the GEN class. Combining Proposition F with Theorem GROW's class (fixed Gaussian designs with growing relevance sets, iid AR(1)) would additionally need C-DLV's Lemma S on that class; this has not been checked and is **not claimed**. (5) The C-ENG implementation of the fallback (behind a switch) has not been audited against this statement by this writer.

7 Implementation status and method requirements

LS0 and the e4 tie fallback were not in the production core; the Wave-1C protocol allowed proofs to fix parameters of these not-yet-shipped components as method requirements, and forbade changes to shipped components (reported as findings instead) [W1C/PROTOCOL.md:40-47]. C-ENG merges them on branch `w1c/ls0-merge` (worktree; not read by this writer, who runs no git commands).

Requirement	content and reason	source	state (HQ, 2026-11-09)
LS0 merge (OP-LS0)	one §6.10 local search from the start mode after the final search (and the c_4 search), centre emitted, no uniforms, provenance `ls0`, GUARD-W semantics	SoP §6.10 / METHOD #8	merged on `w1c/ls0-merge` (C-ENG)
e4 → e4′	LS0 centre = coded `start_mode(st,cap).S` (or a routine returning the unique maximiser when unique) as a data function on every branch; **no consistency result needs the lexicographic tie rule or the truncation fallback**; GROW needs only "some maximiser"	C-PACK MR1–MR3 [W1C/C-PACK/METHOD_REQUIREMENTS.md:6-16]	premise replacement (patches W1C/C-PACK/patches/P_e4_consumers.md)
TDP-IMPL	the e4 tie fallback is W1-N3's reference algorithm TDP, $O(p\sum_b(2^{\lvert C_b\rvert}+(K+1)^2))$ operations; needed only for the delivery resource bound	C-DLV A.2 [W1C/C-DLV/METHOD_REQUIREMENTS.md:8-15]	merged (C-ENG)
MR1 (GROW): LS0 cap	$\ell^{\mathrm{LS0}} = \ell_n + \lvert\tilde{S}_0\rvert$ (final search keeps ℓ_n); any cap $\geq \lvert\tilde{S}_0\rvert$ is accepted by the proof; a cap $o(\lvert\tilde{S}_0\rvert)$ cannot carry the LS0 descent guarantee when $\lvert\tilde{S}_0 \setminus S_0\rvert \to \infty$	[W1C/C-GROW/METHOD_REQUIREMENTS.md:1-25]	on branch (C-ENG)
Calibration fallback (shipped component)	freeze the last successful tier-1 construction or the completed tier-1 pilot schedule instead of returning `schedule_limit` (Prop. 6.3, A1's total-fallback form)	C-DLV B [W1C/C-DLV/METHOD_REQUIREMENTS.md:17-25]	**adopted by HQ** as a method change, behind a switch

C-ENG report. At the time of writing `Theory_work/Wave1C/C-ENG/REPORT.md` is on the C-ENG branch and its summary had not reached the writer; this section must be amended with it (tests, regression,

the order of LS0 among the searches, the switch name and default). Nothing in this chapter asserts that the merged code satisfies OP-LS0, e4′, TDP-IMPL or GUARD-W; these remain named premises, with the integration replay (N2(c)/ENG02) open.

7.1 Integration lemma: the larger LS0 cap

PXN, GEN and C-MIS Cor. D were proved with LS0 cap ℓ_n; C-GROW and Proposition F use $a_n = \ell_n + |\tilde{S}_0| \geq \ell_n$. The lemma below shows which conclusions transfer.

Setting. For a finite-integer-valued deterministic measurable function $a = a(X, y) \geq \ell_n$ of the admissible data (the coded cap is an `int`), let $\mathfrak{M}(a)$ be the method that differs from the specified v3.1 + LS0 only in LS0's move cap, which is a. Structural facts used, each from the specification or the code:

(L1) LS0 is the last search: it runs after the final local search and after the c_4 search ($R' > 4$), and only the HPM running best, the stores, flags and archives are updated after it [SoP/02_method.tex:490-494].

(L2) LS0 reads no uniforms and alters no particle; its centre $\tilde{S}_0$ is the e4′ data function computed before LS0 and does not depend on the cap.

(L3) LS0 executes `local_search(centre, ell, …, emit_centre = true)`: the centre is evaluated; each loop iteration completes a scan of every feasible flip and swap of the current centre C, selects the best scanned neighbour (largest F, exact ties lexicographic), and then: if its F is not strictly larger than $F(C)$, terminates with `no_improvement`, `local_certified`, returning C; otherwise, if the move count has reached `ell`, terminates with `local_limit`, returning that neighbour; otherwise moves to it. The cap enters *only* in this test [CORE/cpp/v3_engine.h:385-425]; same code in [PROD/cpp/v3_engine.h:378-418].

(L4) EX-A (scores exact; F a fixed function on Ω); GUARD-W (on $\mathcal{O}$ the delivered record equals the potential record); the reported HPM is the $\succ$-maximum of the all-scored set.

Let $N(C)$ be the feasible flip/swap neighbourhood of C, and let $C_0 = \tilde{S}_0, C_1, \ldots$ be the *uncapped* greedy path (C_{k+1} = best scanned neighbour of C_k when it strictly improves F); m^* = the number of its moves. Since F strictly increases along the path and Ω is finite, $m^* < \infty$.

Lemma 7.1 (CAP: LS0-cap monotonicity)**.** *Let a, a' be finite-integer-valued measurable functions of the admissible data with $\ell_n \leq a \leq a'$; couple $\mathfrak{M}(a)$ and $\mathfrak{M}(a')$ through the same valid seed and ideal uniforms. The deterministic LS0 search is defined on every admissible data set, also on branches that do not execute it (exact-branch convention); its scored set is distinguished from the set actually scored by the fit. For every data set, seed and uniform realisation:*

(a) Prefix. *LS0 with cap a makes $\min(a, m^*)$ moves along $C_0, C_1, \ldots$, scans exactly $C_0, \ldots, C_{\min(a,m^*)}$, and its scored set is $\mathcal{S}_{\mathrm{LS0}}(a) = \{C_0\} \cup \bigcup_{k \leq \min(a,m^*)} N(C_k)$. It terminates certified at C_{m^*} iff $m^* \leq a$; otherwise with `local_limit`. Hence $\mathcal{S}_{\mathrm{LS0}}(a) \subseteq \mathcal{S}_{\mathrm{LS0}}(a')$, and if the cap-$a$ search ends certified, the cap-a' search is identical (same moves, scans, scored set, return value and flags).*

(b) Potential run. *The potential runs of $\mathfrak{M}(a)$ and $\mathfrak{M}(a')$ coincide up to LS0, $\bar{\mathcal{O}}$ does not depend on the cap, $\bar{\mathcal{S}}^+(a) \subseteq \bar{\mathcal{S}}^+(a')$, $\{LS0(a') \text{ misses } S_0\} \subseteq \{LS0(a) \text{ misses } S_0\}$, and $D_n(a') \leq D_n(a)$.*

(c) Reported HPM. *For $\mathfrak{M}(a)$ with any $a \geq \ell_n$: $\mathcal{O}(a) \cap \{\hat{S}^+(a) \neq S_0\} \subseteq \{\pi(S_0) \leq \frac{1}{2}\} \cup \big(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+(\ell_n)\}\big)$, hence*

$$\mathbb{P}\big(\mathcal{O}(a) \cap \{\hat{S}^+(a) \neq S_0\}\big) \leq 2\,\mathbb{E}_0[1 - \pi(S_0)] + D_n(\ell_n) \leq 2\,\mathbb{E}_0[1 - \pi(S_0)] + \mathbb{P}(LS0(\ell_n) \text{ misses } S_0).$$

Proof. (a) Induction on loop iterations using (L3). Before the $(k+1)$-st scan both runs are at the same centre C_k with the same move count k (true for $k = 0$: the same emitted centre, (L2)). The scan, the selected neighbour and the strict-improvement test depend only on C_k and F ((L4)), not on the cap. If there is no strict improvement ($k = m^*$), both terminate certified at C_k. Otherwise

the cap-a run terminates iff $k \geq a$, i.e. $k = a$ (the count never exceeds the cap); while $k < a$ both move to C_{k+1}. So the cap-a run scans $C_0, \dots, C_{\min(a,m^*)}$ and the scored set is as stated; the cap-a' run scans a superset of centres. If $m^* \leq a \leq a'$ both stop at the same iteration with the same outputs. (b) By (L1)–(L2) everything before LS0 in the potential run uses the same uniforms and the same deterministic computations for both caps, and LS0 changes neither particles nor the calibration; the potential status (`exact`, `completed` or `schedule_limit`) is fixed before LS0, and a capped LS0 returns output, so $\bar{\mathcal{O}}$ is the same event. On a completed SMC potential run, $\bar{\mathcal{S}}^+(a) = \bar{\mathcal{S}} \cup \mathcal{S}_{\mathrm{LS0}}(a)$ with the same pre-LS0 set $\bar{\mathcal{S}}$, so (a) gives the inclusions. On the exact branch every feasible model is already scored and the sets coincide for both caps. On a `schedule_limit` branch neither fit executes LS0, the actual scored sets coincide, and the branch is outside $\bar{\mathcal{O}}$. Thus the inclusions hold on every branch, and $D_n(a') = \mathbb{P}(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+(a')\}) \leq D_n(a)$. (c) Actual output implies potential output ($\mathcal{O}(a) \subseteq \bar{\mathcal{O}}$, the guard only stops), and on $\mathcal{O}(a)$ the delivered all-scored set is $\bar{\mathcal{S}}^+(a) \supseteq \bar{\mathcal{S}}^+(\ell_n)$ (GUARD-W, (b)). If S_0 lies in it and $\pi(S_0) > \frac{1}{2}$ then $F(S_0) > F(S)$ for every $S \neq S_0$, so the $\succ$-maximum is S_0. This gives the inclusion; Markov's inequality gives $\mathbb{P}(\pi(S_0) \leq \frac{1}{2}) \leq 2\mathbb{E}_0[1 - \pi(S_0)]$, and $D_n(\ell_n) \leq \mathbb{P}(\mathrm{LS0}(\ell_n)$ misses $S_0)$ is the operational closure [OP] of §5.1. □

Flagged for review — Lemma CAP

Status: PROVED (by this seat); cross-family check by the fidelity reviewer F1 (Codex gpt-6.1-sol, high): VERIFIED-WITH-AMENDMENTS (finite integer caps; explicit bookkeeping for the exact and `schedule_limit` branches; GEN transfer wording), all applied [W1C/FINAL/reviews/F1_fidelity_sol.md:1-12]. Not councilled. The proof reads the coded `local_search` (frozen core and `PROD/`) and takes (L1)–(L2) from the specification; it does not inspect C-ENG's merged LS0 (in particular that LS0 is still the last search there, and that the MR1 cap is passed as `ell` to the same routine). HQ audit A1 independently states the prefix property [W1C/HQ_AUDIT/A1_propF_redteam.md:56]. Request: council scrutiny of (a)–(c) and of the list below, and a C-ENG confirmation of (L1)–(L3) on the merged code.

What survives under MR1 (cap $a_n = \ell_n + |\tilde{S}_0|$). By Lemma 7.1(c), every conclusion that was obtained by bounding $\mathbb{P}(\mathrm{LS0}(\ell_n)$ misses $S_0)$ or $D_n(\ell_n)$ and $\mathbb{E}_0[1 - \pi(S_0)]$ holds verbatim for $\mathfrak{M}(a_n)$, with $\mathcal{O}$, U_n those of $\mathfrak{M}(a_n)$:

- PXN (C) and (T)'s first clause ("LS0 scores S_0"); (T)'s second clause ("returns S_0 with `local_certified`") too, because on that event the cap-ℓ_n search ends certified and (a) makes the cap-a_n search identical; PXN (K)'s *on-event* clause ($c + 1$ scans on $\mathcal{T}_n$) for the same reason.
- GEN (C2) consequences, (C3), (C3-fav), (C4) and the instances (C5)(a)–(d): these retain their hypotheses and onsets; whenever $\ell_n \geq 2^{\bar{\nu}} - 2$ (part of GEN's onset), also $a_n \geq 2^{\bar{\nu}} - 2$, so the discovery conclusion transfers; more directly, the capped search scores S_0 when $a_n \geq \tau(C) - 1$ and has certified endpoint S_0 when $a_n \geq \tau(C)$.
- C-MIS Corollaries D and D′ (through GEN); Corollary H holds for every cap as stated. M5(b1-LH) (LS-score: LS0 scores S_0 within $\max\{\Phi(S^\circ) - 1, 0\}$ moves) — with its own statuses unchanged.
- C-DLV Corollary D′ and the consistency clause of Proposition F ($\mathcal{K}_n(a_n) \supseteq \mathcal{K}_n(\ell_n)$); SL2's discovery input ($\mathbb{P}(\mathcal{O} \cap \{S_0 \notin \mathcal{S}^+\}) \to 0$); M4(c) retention is a property of the update rule $\mathrm{top}_c(\mathcal{T}_{\mathrm{old}} \cup \mathcal{S}_{\mathrm{added}})$, not of the cap.

What does not transfer verbatim.

- *Worst-case costs.* PXN (K)'s first clause becomes $\leq 1 + (a_n + 1)\max_{k \leq K}[p + k(p - k)] \leq 1 + (\ell_n + K + 1)(p + p^2/4)$ scores — of order $(\ell_n + K)p^2$, not $O(p^2 \log(n + p))$ in general. C-DLV's Lemma R row "LS0 … cap $\ell_n = \ell$", its score count, work/memory envelopes and Corollary R's allocation must be replaced by A1's W_G, M_G and amended allocation (built into Prop. 6.3); the old declared allocation is not shown to prevent a guard stop [W1C/C-DLV/lemmas/L1_resources.md:24-25]. M8's accounting (card TH08) counts local searches with $\ell = \lceil \log_2(n + p + 1) \rceil$ [SoP/04_theory.tex:708]; its LS0 term would have to be re-charged with $a_n + 1$ scans.

- *Flags and metadata.* When the cap-ℓ_n search ends with `local_limit`, the cap-a_n search continues: `local_limit`, the LS0 move count and return value, `ls0_improved`, the `local_certified` flag of the finally reported model, the provenance counts, the score-event ledger, the store contents beyond S_0 (the update top_c can evict models, so stores are not monotone in the cap) and the near-tie list may all differ. No theorem of this chapter asserts these outside the events named above.
- *Negative statements.* GEN (C6) ("P(LS0 from $\tilde{S}_0$ misses S_0) $\to 1$" on its explicit design) is about the cap ℓ_n: its proof uses that LS0 reaches only models within symmetric-difference distance $2(\ell_n + 1)$ of $\tilde{S}_0$. Monotonicity runs the other way, so (C6) is **not established for MR1**; under MR1 the reachable distance is $2(a_n + 1) \geq 2|\tilde{S}_0|$.
- *The actual no-output event.* U_n depends on the cap through runtime (a longer LS0 can meet a guard stop). Joint-with-output statements are stated for each method's own U_n; delivery requires the amended allocation (Prop. 6.3).

8 Scope and frontier

Not claimed / not proved — what the paper must not claim

1. **Selection consistency is asymptotic.** No quantitative finite-n selection-consistency guarantee is proved, in particular none for the study. (Deterministic inequalities such as SL0, M5(0) and Lemma CAP, the sampler identities N1/Corollary E, and the ideal-machine delivery statement of Proposition 6.3 keep their stated finite-n scope.) Onsets: the TH09 binding condition first holds near $n \approx 2 \cdot 10^6$ for the exhibited families (an illustration, not a universal onset) [SoP/04_theory.tex:811]; the LH-iid onset of $L'_n\delta_n \leq \lambda_0/2$ is $\approx 10^{10}$ for typical constants (C-PACK council) [W1C/C-PACK/lemmas/I2_LHiid_unconditional.md:80-81]; PXN's proven margins are vacuous at $n = 200$; GEN's visibility margins at the study size are 0.16–0.94 (< 1); GROW's asymptotic events fail at tested n and S5 needs $\log n \gtrsim 96$. **The study ($n = 200$) is not covered** by any asymptotic result; there, M5(0)'s finite-n composition bracket (with the posterior-miss probability measured, not bounded) is the account.
2. **Polynomial dimension only**: $p \leq n^A$ for fixed A; no uniformity in growing A; exponential p (Song–Liang Cor. 1 regime) is open (CJ18).
3. **Growing relevance sets of factor type**: S5 / common-factor designs with growing p and random-design S5 are not proved (GROW (C6)); obstacles: the sandwich needs $|Q| = O(\log n)$ and NA needs a factor-aware direction count [W1C/C-GROW/REPORT.md:86-90]. Growing s_0 is not covered anywhere.
4. **Delivery for the shipped calibration** is open: event (O) of Theorem 6.1. Delivery is proved only for the fallback calibration of Proposition 6.3, under IDEAL-W, MACH, CONFIG and the amended allocation, on LH, LH-iid and the GEN class (not checked for the GROW class).
5. **Reported pooled PIPs**: no in-probability sure-screening theorem for the shipped $R' = 4$ pooled PIPs; only the islandwise bound, M1(i) under stacking ((h), (i): PROVED (by this seat, unchecked)) and the store-restricted SL2 (§4.2).
6. **Misspecification**: exact-model concentration only under GR (fixed τ^2, fixed projected sparsity, T03♯ hypotheses on the projected truth); EP with growing p_C and small residual, approximate projection, non-Gaussian/heteroscedastic omitted parts, random-design EP and n-dependent τ_n^2 are open; no density-contraction analog of Song–Liang Thm 4.
7. **Hellinger/predictive contraction and exponential rates** (Song–Liang Thms 1–2): not attempted (CJ20).
8. **Memory.** The memory envelope proved sufficient for delivery is $O((n+p)^2 + \text{Cert}_M)$ words

(C-DLV Lemma R; with MR1, A1's M_G, same order); the shipped scorer is Gram-based (a Gram-free scorer is a declared, unshipped v3.2 item [SoP/02_method.tex:414-415]), so quadratic memory in p is a practical limit for $p \gg n$; no sub-quadratic bound is proved.

9. **Floating point.** Every theorem assumes EX-A (exact arithmetic, an idealisation). The only binary64 statement is Lemma UM-FP for the start-mode identification step, under FP-TAB, untested for the coded scorer; delivery also assumes IDEAL-W (the checked-overflow status `unsupported_size` fires for any fixed word size at large enough (n, p)).
10. **Implementation premises** remain premises: GUARD-W audit, integrated-LS0 replay, worker equivalence, IND-T extension ($R' > 4$), N9 START/KERNEL (candidate). LS0 and the fallback are on a branch, not in the frozen core. Song–Liang's Theorem S hypotheses were not inspected.

Recommended next steps (from the ticket reports). (i) Council scrutiny of Lemma 7.1 and a C-ENG confirmation of (L1)–(L3) on the merged code; (ii) extend C-DLV's Lemma S to GROW's class so that Proposition 6.3 covers S2; (iii) GROW's frontier: NA with a low-rank common component and a polynomial ladder for factor designs; a sharper NA restricted to the path for finite-n onsets; (iv) C-MIS: the sufficient-statistic reduction for EP with small residual; (v) route (O-a)/(O-b) for the shipped calibration (pilot stage count, CJ06); (vi) HQ registry updates: N1 card status, TH161/TH163 rows, the patches of C-PACK.

9 Provenance

Council scores are (Codex gpt-6.1-sol / Claude Opus 5.5), round in which both seats passed with all gates. Review files are in each ticket's `reviews/`. Commits are on `main` (messages checked in the local reflog).

Result	proof files	review trail	council	commit
S-T / T03♯ (Thm 3.1)	SoP card TH09; `NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md`	SF W2-VT03R, XF A-REV2 (archive)	— (cited card)	frozen `21ebc28`
SL0, SL1, SL1′, SL2, map v2	`W1C/C-PACK/lemmas/I4_SL_corollaries.md`, `SONG_LIANG_MAP.md`	R34 (ERROR as written → 9 fixes), R38 CONTINUE, R39 sweep, C1	9.0 / 8.7 (r1)	`3b1d795`
e4′, Lemma UM, UM-FP	`W1C/C-PACK/lemmas/I1_e4_replacement.md`, `patches/P_e4_consumers.md`	R30 lock, R33 GAP → 7 fixes, R38, R39, C1	9.0 / 8.7 (r1)	`3b1d795`
LH-iid unconditional (Lemma I2)	`W1C/C-PACK/lemmas/I2_LHiid_unconditional.md`, `patches/P_GEN_LHiid.md`	R35 V-W-A (3 fixes), R38, R39, C1	9.0 / 8.7 (r1)	`3b1d795`
SEL, L8 (C1′) amendments	`W1C/C-PACK/patches/P_SEL_L6d.md`, `P_L8_certificate.md`	R31, R32 (xhigh) V-W-A; R37 re-check	9.0 / 8.7 (r1)	`3b1d795`
PXN consolidated supplement	`W1C/C-PACK/item5_PXN_consolidated/` (00–06, INDEX, MAP, APPLIED_ERRATA)	R36 fidelity FAITHFUL-WITH-FIXES; R39	(item 5 outside council scope)	`3b1d795`
Theorem PXN	`W1/W1-PXN/FINAL_PROOF.md`, `W1/W1-PXN/lemmas/`	R00 lock; R11–R17, R19, R21 item checks; R20 CONTINUE; R30 sweep (1 ERROR, 2 GAP, 4 MINOR fixed)	8.6 / 8.3 (r1)	`f1751f3`
W1-N3 (start mode as coded, TDP)	`W1/W1-N3/PROOF.md`, `W1/W1-N3/lemmas/`	R01 lock; R02, R03, R05 checks; R04 CONTINUE; R06 sweep	9.3 / 8.6 (r1)	`21d5e80`

Result	proof files	review trail	council	commit
Theorem GEN	`W1/W1-GEN/lemmas/00_MAIN.md`, L1–L8	R01 lock; R02–R04, R06–R09 checks; R05 CONTINUE; R10 sweep (0 fatal)	9.0 / 8.9 (r1)	`94ac305`
Lemma N1, Cor. E	`W1/W1-N1/lemmas/N1_theorem.md`, `E_corollary.md`, L1–L3	SL1 lock; LC1; MP1 CONTINUE; SW1 sweep (0 critical)	9.3 / 8.9 (r1)	`f16ce15`
Theorem MIS, F, Cor. H/D/D′, Lemma Z	`W1C/C-MIS/lemmas/00_MAIN.md`, L1–L6	R01 lock; R02, R03, R05 checks; R04 CONTINUE; R06 sweep (0 fatal)	9.2 / 9.0 (r1)	`05dcd85`
Theorem D, Cor. D′, (O), Prop. F (C-DLV form)	`W1C/C-DLV/lemmas/L1_resources.md–L5_insertions.md`, `INVENTORY.md`	R01 REJECT → v2; R02–R04 checks; R05 CONTINUE; R06 sweep; C1 not passed (7.6 / 8.6), fixed	9.0 / 8.9 (r2)	`604f58a`
Prop. F (corrected, Prop. 6.3)	`W1C/HQ_AUDIT/A1_propF_redteam.md`	HQ red team (Codex Sol xhigh): HOLDS-WITH-AMENDMENTS	— (not councilled)	`af2c454`
Theorem GROW	`W1C/C-GROW/lemmas/00_MAIN.md`, L0–L6	R01 lock; R02–R04, R06 (Opus), R07 checks; R05 CONTINUE; R08 sweep (0 fatal, 0 major)	9.2 / 9.0 (r1)	`1853fcd`
Lemma CAP (Lemma 7.1)	this chapter, §7.1	F1 fidelity reviewer (Codex Sol high) VERIFIED-WITH-AMENDMENTS, applied; A1 states the prefix property independently	—	—

Sources: [W1C/C-PACK/REPORT.md:66-82]; [W1/W1-PXN/REPORT.md:71-82]; [W1/W1-GEN/REPORT.md:71-83]; [W1/W1-N1/REPORT.md:67-86]; [W1/W1-N3/REPORT.md:52-61]; [W1C/C-MIS/REPORT.md:116-129]; [W1C/C-DLV/REPORT.md:74-86]; [W1C/C-GROW/REPORT.md:72-82].